

Preventing the Spread of Airborne Pathogens using Critical Area Risk Assessment

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**ΤΜΗΜΑ ΜΗΧ. Η/Υ & ΠΛΗΡΟΦΟΡΙΚΗΣ
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Ευχαριστίες

Θα ήθελα να εκφράσω την ειλικρινή μου ευγνωμοσύνη και να αφιερώσω την παρούσα εργασία στην οικογένειά μου, η οποία με στήριξε αδιάλειπτα σε κάθε βήμα της πορείας μου, παρέχοντάς μου δύναμη, υπομονή και ηθικό στήριγμα καθ' όλη τη διάρκεια των σπουδών μου. Ιδιαίτερες ευχαριστίες οφείλω στον επιβλέποντα Καθηγητή μου, κ. κύριο Σταύρο Δ. Νικολόπουλο, του Τμήματος Μηχανικών Ηλεκτρονικών Υπολογιστών και Πληροφορικής του Πανεπιστημίου Ιωαννίνων, για την τιμή που μου έκανε αναλαμβάνοντας την επίβλεψη της παρούσας διπλωματικής εργασίας. Η καθοδήγησή του, η επιστημονική του υποστήριξη και οι εύστοχες παρατηρήσεις του υπήρξαν καθοριστικές για την ολοκλήρωση του παρόντος έργου. Θερμές ευχαριστίες απευθύνω επίσης και στα μέλη της τριμελούς εξεταστικής επιτροπής αλλά και στον Επίκουρο Καθηγητή κ. Ιωσήφ Πολενάκη, του Τμήματος Πληροφορικής του Ιονίου Πανεπιστημίου, για την πολύτιμη βοήθειά του και την πρόθυμη υποστήριξή του καθ' όλη τη διάρκεια της ερευνητικής διαδικασίας. Του εύχομαι κάθε προσωπική και επαγγελματική επιτυχία. Ακόμη, Θα ήθελα επίσης να ευχαριστήσω θερμά τον κ. Χρίστο Χόνδρο, για τη συμβολή του στην ανάπτυξη της βάσης του μοντέλου προσομοίωσης, καθώς η μελέτη του αποτέλεσε θεμέλιο λίθο πάνω στον οποίο βασίστηκα προκειμένου να επεκτείνω, τροποποιήσω και να ενσωματώσω τη δική μου ερευνητική εργασία.

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Συγγραφέας

Μαρκάκης Φώτιος

Περίληψη

Στην παρούσα εργασία μελετώνται οι παράγοντες που συμβάλλουν στην εξάπλωση αερομεταφερόμενων παθογόνων και προτείνεται ένα ολοκληρωμένο αλγοριθμικό πλαίσιο πρόληψης επιδημιών. Λαμβάνοντας υπόψιν κρίσιμες παραμέτρους μετάδοσης του ιού SARS-CoV-2, αναπτύχθηκαν τρία υπο-μοντέλα: το χωρικό, το μοντέλο κίνησης και το μοντέλο μετάδοσης. Το χωρικό μοντέλο μετατρέπει πραγματικές χαρτογραφικές εικόνες πόλεων σε γραφήματα που αποτυπώνουν τη δομή του αστικού ιστού. Το μοντέλο κίνησης εφαρμόζει στοχαστική agent-based προσέγγιση για την προσομοίωση των μετακινήσεων των πολιτών, ενώ το μοντέλο μετάδοσης ενσωματώνει τόσο το επιδημιολογικό όσο και το φυσικό υπόβαθρο της εξάπλωσης του ιού. Στο υπάρχον πλαίσιο προσθέσαμε έναν μηχανισμό αποτίμησης επικινδυνότητας κρίσιμων περιοχών, ο οποίος επιτρέπει τη δυναμική αξιολόγηση του κινδύνου έκθεσης σε τοπικό επίπεδο. Συγκεκριμένα, γύρω από κάθε άτομο ορίζεται μια τοπική ζώνη αναφοράς, μέσα στην οποία καταγράφεται η παρουσία και η επιδημιολογική κατηγορία των γειτονικών ατόμων. Βάσει αυτών των πληροφοριών υπολογίζεται ένας δείκτης επικινδυνότητας $A(r)$, που αποτυπώνει ποσοτικά την ένταση του επιδημιολογικού κινδύνου σε συγκεκριμένες περιοχές. Στη συνέχεια, ένα ποσοστό του πληθυσμού που έχει χαρακτηριστεί τυχαία ως «υπάκουο» ανταποκρίνεται στις ειδοποιήσεις του συστήματος: αν εντοπίσει ότι βρίσκεται σε κρίσιμη ζώνη, αλλάζει διαδρομή ή επιστρέφει στην αφετηρία του, μειώνοντας έτσι τις πιθανότητες έκθεσης. Οι «ανυπάκουοι-ασυνείδητοι» πολίτες συνεχίζουν τις προγραμματισμένες τους διαδρομές, αγνοώντας τις προειδοποιήσεις. Η υλοποίηση αυτή επιτρέπει την προσομοίωση όχι μόνο της εξάπλωσης του ιού αλλά και της αλληλεπίδρασης ανάμεσα στον ατομικό βαθμό συμμόρφωσης και στη χωρική κατανομή του κινδύνου. Μέσω αυτής της προσέγγισης, ενισχύεται η δυνατότητα έγκαιρης πρόληψης της εξάπλωσης παθογόνων, καθώς καθίσταται εφικτός ο εντοπισμός και η αποφυγή επικίνδυνων περιοχών σε πραγματικό χρόνο. Η αποτελεσματικότητα του συστήματος αξιολογήθηκε μέσω επαναλαμβανόμενων προσομοιώσεων τύπου Monte Carlo.

Λέξεις Κλειδιά: Γραφήματα, Αλγόριθμοι, Αποτίμηση επικινδυνότητας περιοχών, Αποφυγή Πανδημιών, Προσομοίωση

Abstract

In this work, we study the factors that contribute to the spread of an airborne pathogen and propose a comprehensive algorithmic framework for epidemic prevention. Considering critical transmission parameters of the SARS-CoV-2 virus, three sub-models were developed: the spatial model, the mobility model, and the propagation model. The spatial model transforms real cartographic images of cities into graphs that represent the structure of the urban environment. The mobility model applies a stochastic agent-based approach to simulate the movements of citizens, while the propagation model incorporates both the epidemiological and physical foundations of viral transmission. Within this framework, we introduced an Area Risk Assessment mechanism, which enables the dynamic evaluation of exposure risk at the local level. Around each individual, a reference zone is defined, within which the presence and epidemiological category of neighboring individuals are recorded. Based on this information, a risk index $A(r)$ is computed, quantitatively expressing the intensity of epidemiological risk in the corresponding micro-environment. A fraction of the population, randomly designated as *obedient*, responds to system alerts: when they detect that they are located within a critical zone, they either change their route or return to their origin, thereby reducing their likelihood of exposure. Disobedient citizens continue to follow their scheduled routes, ignoring the warnings. This implementation makes it possible to simulate not only the spread of the virus but also the interaction between individual compliance levels and the spatial distribution of risk. Through this approach, the framework strengthens the ability for timely prevention of pathogen spread, as it enables the identification and avoidance of high-risk areas in real time. The effectiveness of the system was evaluated through repeated Monte Carlo simulation experiments.

Keywords: Graphs, Algorithms, Area Risk Assessment, Pandemic Prevention, Simulation

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Section 1. Introduction

In this section, we undertake a comprehensive examination of the foundational principles of epidemiology, explore significant contributions from key research in the field of pandemic prevention, and provide a concise overview of the strategic framework guiding the present study.

1.1 Epidemiology

Epidemiology, as a foundational discipline within the health sciences, is concerned with the investigation of how diseases originate, propagate, and persist within populations. It seeks to quantify patterns of disease occurrence and distribution while systematically identifying underlying determinants and risk factors that inform strategies for prevention, control, and public health policymaking [1–4]. Among the core analytic methodologies employed in this field are compartmental models, which divide the population into distinct categories or states, such as susceptible, infected, infectious, and immune, and mathematically characterize the transitions between these states. These modeling approaches enable the estimation of transmission dynamics and the assessment of intervention effectiveness. A more detailed discussion of these models is presented in the sections that follow.

1.2 Pandemic Prevention

Preventing the escalation of an infectious disease into a pandemic, where a large portion of the population becomes infected, requires the timely adoption of multiple countermeasures. A pandemic may emerge on either a local or global scale, depending on the size and susceptibility of the population at risk. Manika and Golden [29], in their study of the H₁N₁ influenza pandemic, argue that proactive prevention is often more effective than reactive mitigation of its consequences.

Many pathogens reach humans through zoonotic transmission pathways [30], and through mutation they adapt to human-to-human transmission, thereby fueling pandemics. As highlighted in [28], the human–animal interface plays a critical role in this process, making its study essential for the development of preventive strategies. Morse et al. [31] emphasize that mathematical modeling and information technologies can support early identification of emerging pathogens and provide more accurate risk assessments. Similarly, Jung et al. [32] focus on influenza and demonstrate the significance of time-dependent interventions, particularly quarantine, as a key strategy for pandemic prevention.

Historical pandemics, such as the Great Plague of London (1665–1666) and the Spanish flu (1918–1920), resulted in millions of deaths, often in the absence of adequate prevention measures. Today, with abundant computational resources and digital data, the scientific community is better equipped to predict, prevent, and manage pandemics. Nevertheless, the emergence of the SARS-CoV-2 virus in 2019, a novel strain of the Coronavirus, developed into a global pandemic despite these advancements. As outlined in Shahzad et al. [33], the rapid implementation of countermeasures in highly congested areas is crucial to curtail transmission. Finally, Stawicki et al. [34] stress the importance of integrating diverse COVID-19 data sources to strengthen pandemic readiness, while also pointing to key gaps in global awareness revealed during the crisis.

1.3 Related Work

Next, we review the most relevant studies that employ similar modeling approaches for simulating pandemics. These include works that propose epidemic models or simulation frameworks, as well as research focused on simulating individual contact dynamics, particularly in relation to the physical principles underlying the transmission of airborne viruses.

Currie et al. [5] synthesize the state of practice on how simulation and modeling inform responses to SARS-CoV-2/COVID-19, emphasizing their usefulness for decision support and pandemic prevention. Cliff et al. [6] proposes an agent-based framework for influenza that analyzes spatiotemporal spread using nationally sourced mobility patterns an approach that resonates with our focus on movement-driven transmission dynamics—and evaluate both quantitative and qualitative spatiotemporal profiles. At the global scale, Hufnagel et al. [7] formulate a stochastic scheme with sub-models for infection and transmission dynamics to explore worldwide control.

Kovačević et al. [8] introduce a distributed, continuous-time epidemic model that incorporates social distancing via statistical descriptors rather than classic compartmental transitions, enabling combined interventions with minimal changes to govern equations and accommodating longer incubation periods typical of COVID-19. Kevrekidis et al. [9] develop the SEIHARD model for Greece and Andalusia, highlighting incubation and asymptomatic spread—elements that align with our emphasis on pre-/asymptomatic phases. Bertacchini et al. [10] stress the temporal signatures of SARS-CoV-2 propagation, arguing that COVID-19’s complex behavior follows spatiotemporal pandemic dynamics and that careful temporal analysis can surface actionable countermeasures.

Methodologically, Becker and Grenfell [11] apply tsIR/time-series SIR to infer incidence and analyze disease time series. Modi et al. [12] estimate COVID-19 progression via SEIR across varying R_0 . Duan et al. [13] review mathematical, network-based, and agent-based models, outlining principles, strengths, and limitations. Nadini et al. [14] use time-varying networks with SIR/SIS to quantify how temporal connectivity influences spread; results are validated on synthetic and real modular/temporal graphs. Barmak et al. [15] study mobility’s role in dengue via a stochastic human–mosquito system, showing how movement patterns modulate epidemic dynamics. Naoki and Holme [16] review temporal contact networks for prediction and control, while Obadia et al. [17] survey methods for estimating transmissibility, implementing five as a toolbox, and compare sensitivities across estimators. Britton [18] contrasts deterministic and stochastic formulations and presents an endemic-disease model. Cuevas-Maraver et al. [19] analyze lockdown impacts in single- and two-age SEIR extensions (with presymptomatic states) for Mexico. Keeling and Eames [20] outline how epidemiology and network theory combine to illuminate disease dynamics. Valdano et al. [21] derives epidemic thresholds bridging discrete/continuous-time in evolving networks; Tizzani et al. [22]

quantify effects of time-varying topology and memory on SIS/SIR dynamics; and Liu et al. [23] examine thresholds on multiplex, time-varying networks under SIS.

Turning to airborne transmission physics, Mark et al. [24] assesses infection risk from respirable pathogen emissions, linking risk to airborne concentration and particle volume. Drossinos and Stilianakis [25,26] integrate droplet aerodynamics into transmission models and detail how turbulent expiratory flows shape exposure. Wei and Li [27] further examine how human activities (e.g., door opening, walking) perturb airflow and droplet transport, altering transmission risk.

Complementary to this corpus, if modeling infrastructure is concerned, Stuart et al. [28] build HPVsim on the Starsim platform, utilizing its modular agent-based architecture to simulate HPV transmission and cervical disease dynamics. The Starsim family is designed for infectious disease applications requiring reproducible scenario design and rapid model reuse. In HPVsim, the platform couples flexible state machines with layered contact structures and configurable policy levers. Abey Suriya et al. [28] adapts Starsim to implement a novel approach for modeling the natural history of HPV independently of specific screening modalities, while Zimmermann and Boudreau et al. [28] enhances its computational performance to support large-scale, high-resolution simulations with reduced stochastic noise.

Cohen et al. [29] employ a dynamic framework to evaluate the changing health impact of vaccines throughout the pandemic, accounting for vaccine efficacy, public health policies, and viral transmissibility across different waves. Their results emphasize the crucial role of timely vaccine deployment and adaptive strategies in mitigating disease burden amid emerging variants. Complementing this, Covasim is an agent-based COVID-19 model that links detailed, layered contact networks—covering households, schools, workplaces, and community settings to within-host disease progression and a comprehensive intervention suite including testing, tracing, isolation, quarantine, vaccination, and behavioral change. Covasim provides built-in calibration, parameter uncertainty quantification, and counterfactual analysis to support robust policy evaluation.

To conclude this section, we single out the work of Chondros, Nikolopoulos, and Polenakis [30] as the most directly relevant and foundational to our study. This model is our baseline. We build on its map-to-graph pipeline, attraction-driven movement, distance-modulated contagion, and emphasis on intervention timing to support localized risk assessment and policy testing.

1.4 Motivation and Contribution

Through recent and recurring global outbreaks of airborne pathogens, including the SARS-CoV-2 pandemic (COVID-19), numerous studies have focused on strategies for epidemic control. Motivated by the substantial number of infection cases leading to ICU admissions and fatalities, our research seeks to advance the intersection of computer science and public health. Specifically, we aim to advance area risk validation as a computational approach to support the suppression of viral spread that might escalate to pandemic scale.

In this work, we study the propagation of airborne viruses and propose a simulation framework that aims to suppress epidemic dynamics before they escalate into a pandemic. The framework is structured around three core components: a spatial model that defines the environment in which the population moves and transmits the virus, a mobility model that simulates the movement patterns of individuals within this environment, and a propagation model that captures the mechanisms of virus transmission between hosts.

As a case study, we focus on SARS-CoV-2, modeling its transmission under various countermeasures such as lockdowns, density reduction, and digital contact tracing. Through a series of simulation experiments, we extract valuable insights into the effectiveness of these interventions, emphasizing the crucial role of response time: early deployment of countermeasures strongly suppresses viral spread, whereas delays may allow outbreaks to develop into pandemic conditions.

Building upon this framework, we extend our study with a dedicated area risk assessment component. This addition investigates how localized conditions and spatial heterogeneity influence infection dynamics, allowing us to identify high-risk zones where contagion is more likely to intensify. By incorporating critical area risk assessment, we are able not only to evaluate the general effectiveness of countermeasures but also to examine their targeted impact on specific locations, providing a more precise and actionable understanding of how to contain airborne pathogens.

In this work, we conduct research on the epidemiological dynamics of airborne virus transmission, focusing on how countermeasures can alter the course of an outbreak and ultimately prevent its escalation into a pandemic. The main contribution lies in the design of an integrated simulation framework that captures the complex behavior of epidemic spread through a hybrid modeling approach. Specifically, the framework incorporates a spatial model that simulates the environment of a city using realistic map images, a mobility model built on agent-based principles that reproduces the interactions and movement patterns of citizens, and a stochastic propagation model that describes viral load development within hosts and governs transmission between individuals.

Beyond these core elements, the framework allows the integration of external algorithms for the implementation and evaluation of diverse countermeasures. This enables us to explore their direct influence on pandemic dynamics and assess their capacity to mitigate mortality rates. Valuable information that potentially could be of significant importance, is the results exhibited by the conducted investigation over the series of Monte Carlo experiments over the proposed countermeasures, regarding their effect on death-rates in terms of numbers of patients that given a standard set of Intensive Care Unit positions is able to avoid potential deaths that potential could not survive if their number exceeds the capacity of ICU positions.

As an extension, we place emphasis on area risk validation experiments, which provide localized assessments of contagion risk within urban environments. These experiments highlight high-risk zones, enabling targeted interventions that not only improve the efficiency of countermeasures but also strengthen early-warning capabilities. The positive outcomes of this approach demonstrate their potential to enhance preparedness strategies and deliver actionable knowledge for public health decision-making.

1.5 Roadmap

An overview of the thesis structure is presented below. In Section 2, we present the theoretical background and the prerequisite knowledge upon which this work is developed. This includes the modeling principles used to simulate the spread of infectious diseases, the specific properties of airborne virus transmission, graph-theoretic approaches for representing contagion among susceptible populations, and techniques that have been applied for pandemic prevention. At the same time, we introduce the integrated simulation framework for pandemic prevention, analyzing its core components: the spatial model, the mobility model, and the propagation model, with reference to the case study of COVID-19. In Section 3, we extend this framework

with our proposed addition: an area risk assessment feature designed to evaluate contagion risk in critical zones. This extension enables localized validation experiments, providing insights into high-risk areas and informing targeted interventions to improve prevention strategies. Section 4 discusses the system’s architecture, outlining its design principles and describing the simulation of uncontrolled virus spread as well as the impact of countermeasures, such as population density reduction, in suppressing transmission. Next in Section 5, we present the experimental study, performing sequential Monte Carlo simulations to evaluate the effectiveness of different countermeasures, including our area risk validation approach. In the final section, Section 6 concludes the thesis with a summary of findings, identification of open problems, and directions for future research aimed at optimizing and expanding the proposed framework.

Section 2. Fundamental Principles

In this section, we present the theoretical foundations and the integrated simulation framework that supports our approach to pandemic prevention and mitigation. We begin by outlining the principles underlying the modeling of spatial parameters, the stochastic mobility patterns of individuals, and the epidemic models used to capture the transmission dynamics of airborne viruses, with particular reference to the COVID-19 pandemic. Building on this foundation, we introduce the proposed framework in detail, analyzing its components, the spatial model, the mobility model, and the propagation model, that together enable the realistic simulation of viral spread and the evaluation of countermeasures.

2.1 Spatial Parameters and Transportation Networks

A key factor influencing the spread of contagious diseases lies in the spatial parameters. High population density and congestion accelerate transmission, particularly in the case of airborne viruses, making this aspect central to our research and its integration into the proposed framework for pandemic prevention. In line with [45], we simulate the underlying city texture by processing real images from Google Maps into graph structures, which define streets and routes for agents moving within the city. Agent-based simulations have been widely adopted in related studies [40–44], as they allow stochastic modeling of community transportation and individual interactions. Such models provide realistic representations of human behavior, making them highly suitable for studying social networks and mobility patterns in the context of epidemic spread [5]. In our framework, we implement a stochastic mobility model that defines probabilities for multiple aspects of individual movement: the frequency of daily routes, destination choices, and whether agents continue or return to their base location. Complementing this, we incorporate a contact graph that represents potential social relationships, enabling linked individuals to remain in close proximity for a period when they meet. This integration enhances the realism of our model and strengthens its ability to simulate airborne virus transmission in urban settings.

2.2 Spread of Air-Born Viruses

In order to model the spread of a disease, and more precisely to simulate its propagation among susceptible individuals and underlying epidemic model is required for the

depiction of the states that an individual may be, and additionally the underlying properties that may be required in order to model the physical substance of the transmission of an airborne virus. Next, we discuss both approaches and further provide an introductory discussion on what we utilize in our proposed approach for the propagation model that is incorporated by our proposed framework.

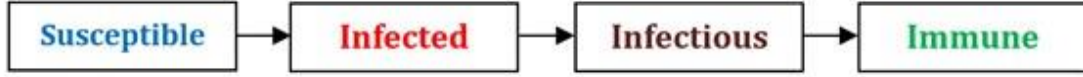


Figure 1. Representation of the states of the SIII epidemic model.

In Figure 1 we present the adopted SIII epidemic model, which extends the classical compartmental frameworks by incorporating four distinct states. This formulation captures the incubation phase before contagiousness, the period of active transmission, and the eventual transition to immunity, aligning with the epidemiological characteristics of SARS-CoV-2 and supporting more accurate simulation of airborne virus propagation.

To formally model the propagation of a spreading disease, there has already been proposed since 1927 by Kermack and McKendrick, the utilization of the so-called compartmental models. Through this modeling, the states of the parts of the population among which a disease is spreading have a specific labeling, namely Susceptible, Infected and Recovered, consisting of the SIR epidemic model, that describes the transition over whose states (S, I, R) an individual may be during the spreading of a disease. That is, modelling the spread of a disease that any individual is Susceptible to the pathogen, then after his infection by the pathogen is labeled as Infected and finally after his treatment or recover turns to the state Recovered, is preferable to model disease that once a Susceptible individual gets the disease, turning for a specific period to the state of Infected, after his treatment the individual is Recovered, been no longer in the Susceptible state, without however the exclusion of being Immune to the disease, that is described by another epidemic model. The SIR epidemic model has several extensions that are adapted to describe more accurately diseases that exhibit more specific behavior. For example, if an epidemic model is required for the description of state transition of influenza, then knowing that an individual may get Infected more than once, the SIS epidemic model may be deployed to describe the behavior of such a disease [35]. In that model the third state corresponds to the state of Susceptible (i.e., Susceptible, Infected, Susceptible, Susceptible) in order to permit the transition from the state of Infected to the state of Susceptible, allowing the probability of getting Infected again. In case of the existence of an incubation period, an extension of the SIR epidemic model, the SEIR epidemic model may be applied to describe the transitions of the population among these states (i.e., Susceptible, Exposed, Infected, Recovered), including also the state Exposed [36], in order to describe the transitional state between the first date of infection and the day that symptoms or any other metrics shows that the individual in Infected, as represented in Figure 1. 11 In our proposed framework for the simulation of the spread of an airborne virus among individuals that move inside the boundaries a of a city area, and taking into account the specifications exhibited by the spread of the SARS-CoV-2 coronavirus we deploy the SIII epidemic model, including the states (Susceptible, Infected, Infectious, and Immune), where an individual primarily is Susceptible to get Infected by the virus, after the exposition to the virus the individual is getting Infected and from a short time interval (i.e., 4-5 days from exposure to symptoms onset [37-39]) and after his starts to be contagious, and hence being

Infectious, while after a period of days he recovers being Immune and not contagious. Since there is not enough information from research in the literature about reported cases of reinfection, at least from the same strain of the SARS CoV-2, we estimate that the SIII model fits perfectly meeting the requirements of our proposed framework. Finally, regarding the second approach that references the physical substance of the transmission of airborne viruses in our approach we developed and propose a model that considers the main median over which the new coronavirus is transmitted, which is the infected droplets.

Throughout this aspect we define two main procedures, namely, the evolution of the viral load inside the infected hosts that consequently defines the percentage of the infected droplets emitted by an infectious host during actions like coughing, sneezing, talking, or even breathing and the spatial method that these droplets are transmitted to susceptible individuals. Hence, having modeled and as close as possible to reality simulated the evolution of the viral load and to a further extent the percentage of the infected emitted droplets, we compute stochastically a probability for an infectious individual to perform an action, e.g., coughing, considering the corresponding volume of emitted droplets. Then, we compute the number of susceptible individuals that are located proximally to this infectious individual and computing a geometrical representation of the produced cloud of droplets in a specific distance we stochastically infect some of the individuals.

2.3 Pandemic Prevention

In our approach, we examine how various countermeasures contribute to the mitigation and, ultimately, the suppression of an epidemic before it escalates into a pandemic. For the purposes of this study, we consider epidemics, specifically the spread of an airborne virus such as SARS-CoV-2, to be classified as a pandemic once 80% of the population becomes infected.

We evaluate a range of countermeasures, including lockdowns that reduce population density in urban areas, self-quarantine of confirmed infected individuals, contact tracing through mobile applications, as well as combinations of these strategies. Using our integrated simulation framework, we conduct a series of Monte Carlo experiments to assess the effectiveness of each measure, averaging results across multiple runs for reliable comparison.

The evaluation focuses on several key factors: the proportion of susceptible individuals that remain uninfected, the total number of infections, the “growth-level” that reflects the aggressiveness of the spread, and the overall duration of the epidemic. We also determine whether the epidemic evolves into a pandemic. An additional finding, aligned with the work of [45], highlights the critical role of response time: even highly effective countermeasures lose impact if applied too late.

Finally, we extend our analysis by estimating potential fatalities based on the WHO-reported mortality rate of 0.3% for SARS-CoV-2. By applying this rate to the daily infection counts produced by our simulations, we compare outcomes both across different countermeasures and against the baseline case where no intervention is applied. This enables us to identify which strategies or combinations are most effective and preferable for deployment.

2.4 Basic Components

The proposed framework for pandemic prevention integrates three sub-models to simulate the spread of an airborne virus within a city environment. These are the spatial model, the mobility model, and the propagation model. The spatial model reproduces the city's topology, capturing its streets and urban layout. The mobility model, implemented as an agent-based approach, simulates the movement patterns of individuals as they travel between specific points in the city. Finally, the propagation model governs the transmission dynamics of the virus, embedding the epidemic model and the specific transmission characteristics of airborne pathogens, here applied to SARS-CoV-2.

2.5 Spatial Model

In this section, we present the transformation of a Google-Map image to an object that depicts the structure of a city, modeling among others the orientations exhibited by the city roads, the points of interest that attract the citizens' routes, and the underlying undirected graph utilized for the construction of the paths in the corresponding routes.



Figure 2. Initial Image from Google Maps

In this figure, we present the original Google-Map image that will be transformed into a structured object that represents the urban layout of the city. The model captures key orientations of the road network.

2.5.1 Transforming G-map Images to Undirected Graphs

The purpose of this operation is a detailed graph-based representation of the image's roads, that model the roads of the city, required for the simulation. At the end of the transformation, an undirected Graph is constructed, whose vertex set correspond to a square spot of the street (i.e., pixel of the image), while an edge belongs to the edge set of the Graph if the corresponding end points are neighboring square spots (i.e., 7-neighborhood) of the street, either left, right, forward, backward, or diagonal. The proposed model is flexible enough, to transform any given image of any size into a Graph alongside with all the auxiliary information required. The purpose of this method is to create a graph whose vertex set corresponds to the streets from the given Google Map image, modeling the area in which the propagation is investigated. More precisely, we selected the preferred area from the Google Maps web application see. Next, we transform the image into a two-dimensional matrix, with each specific cell containing three different values (i.e., the RGB color index of each pixel). To distinguish the cells that correspond to streets and the cells that correspond to buildings, we first transform the initial image into a grey-scale image, casting the RGB index value of each cell into a single value that represents the intensity of gray color, and applying a thresholding procedure, see Figure 3 (b).

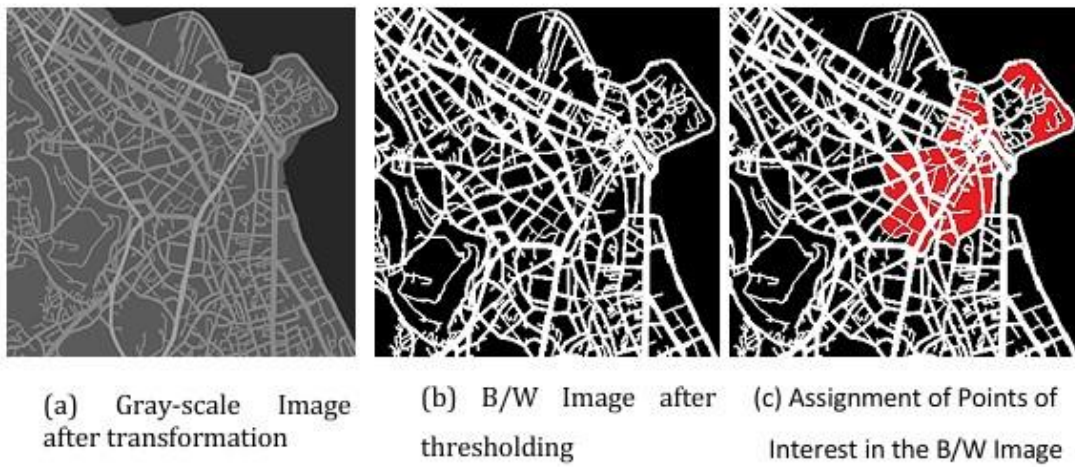


Figure 3. Image pre-processing

In Figure 3 we present the transformation pipeline used to derive a graph-based representation of the city's streets. Beginning with a gray-scale conversion of the original Google Map image, a thresholding step produces a black-and-white rendering that separates streets from buildings and obstacles. Points of interest are then assigned on the processed image, enabling the construction of an undirected graph where vertices correspond to street pixels and edges connecting neighboring cells. This graph forms the foundation of the spatial model employed in the simulation framework.

In our approach we consider that the cell values of the matrix above the specified threshold refer to square spots of streets, while cell values of the matrix below the

specified threshold refer to square spots of the area outside the streets (i.e., buildings, obstacles, etc.). Particularly, we select a threshold and compare it with each cell of the matrix, where cell values above the threshold are casted to value 255 representing the streets, while cell values below the threshold are casted to 0 representing the buildings or any other obstacle. The application of this procedure results in a two-dimensional matrix full of 0 and 255 and a corresponding black-white image that represents the initial image with white streets and black buildings. To this end, it is important to note that the exact value of the threshold is strictly affected by specific characteristics across various images. 16 Considering the two-dimensional matrix full of 0 and 255 the actual graph-construction procedure begins by transforming each particular cell whose value is 255 (i.e., a cell that represents a square spot of the street) into a graph-node. More specifically, our model scans horizontally the cells with value 255 of the two-dimensional structure and creates a G-node with an ID according to specific cell's position. For example, the procedure will create a G-node with $ID = (i,j)$ if and only if the value of the cell in the 25th row and 17th column equals to 255. Alongside the creation of the G-nodes, the same procedure constructs the edges of the graph considering the neighbors of the cells that also have value equal to 255. Two cells are considered as neighboring and their corresponding nodes are joined by an edge, if and only if the cells are adjacent in the matrix regarding the 7-neighborhood. The result is a detailed graph-based representation of the initial image whose vertices represent the actual square spots of the streets, and the edges represent the paths (i.e., hops, or, steps between two adjacent square spots of the street) among them.

2.5.2 Assign Points of Interests

To increase the realism of our simulation, we account for the mobility intensity and traffic patterns typical of a city. Areas of high activity, or points of interest, are zones with frequent movement and congestion, which significantly contribute to the rapid transmission of airborne viruses. At the beginning of each experiment, the user designates these points of interest directly on the initial image, highlighting the high-traffic regions of the city (see Figure 3. (c)). It is worth noting that only the non-street segments (i.e., buildings or obstacles) need to be marked, as these are used to assign an important level reflecting the intensity of traffic in their vicinity. Once this marking is complete, our framework builds the Attraction-Matrix, which models the traffic distribution across the city's roads. Every cell of this matrix corresponds to a square spot of the street and is assigned a value between 0 and 10. Higher values indicate more heavily trafficked cells. Specifically, we define *hot spots* as cells with values in the interval $(7,10]$, *warm spots* as cells with values in $[4,7]$, and *cold spots* as cells with values in $[0,4)$.

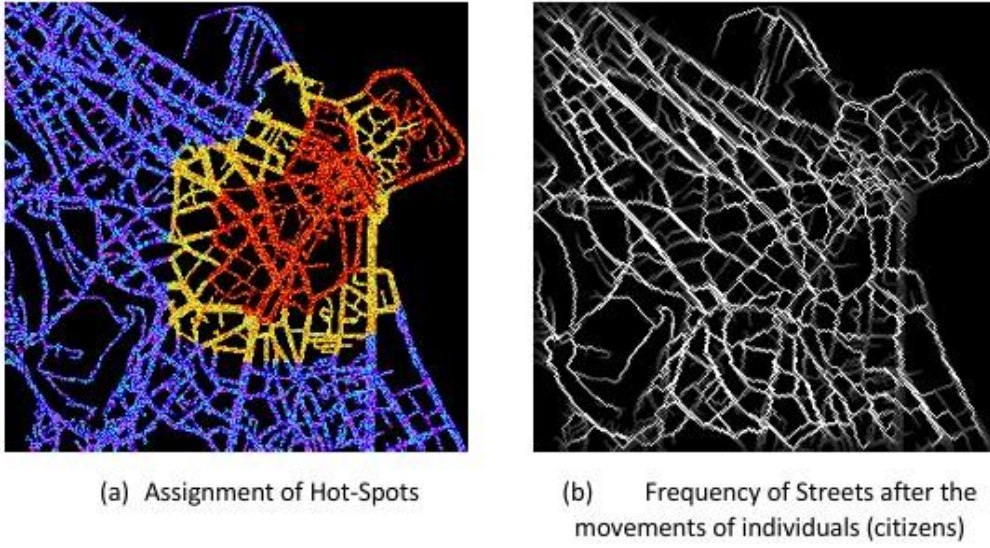


Figure 4. Citizen Mobility with respect to the Hot-Spots

In this figure, we present the assignment of hot-spots within the city map and the resulting frequency distribution of street usage after simulating citizen movements. The classification of zones into hot, warm, and cold spots reflects the varying levels of traffic intensity, while the frequency map highlights how individual trajectories collectively shape congestion patterns. Together, these representations form the Attraction-Matrix, which serves as a foundation for modeling realistic mobility flows and identifying critical areas where viral transmission is most likely to occur.

After the initial points of interest are set, an algorithm propagates their influence across the street network. For each street cell, the closest attraction point is identified, and the corresponding value is assigned. For example, cells located very close to an attraction point are marked as hot spots with values between 10 and 7, whereas distant cells are classified as cold spots with values between 3 and 0. The Attraction-Matrix is a key element of the proposed framework, as it underpins the definition of routes and paths followed by individuals when moving from a source to a destination point in the city. Its role and practical applications will be examined more thoroughly in the next section.

2.5.3 Constructing the Route Basis

In this section, we describe the procedure for constructing the route basis once the attraction points have been defined. As previously discussed, each road is represented as a collection of pixels from the original image, which are mapped onto the corresponding cells of the Attraction-Matrix. Each such cell holds a value in the range $[0,10]$, with $[0,4)$ denoting cold spots, $[4,7]$ warm spots, and $[7,10]$ hot spots. Considering the undirected Graph (see Section 2.5.1), each node corresponds to a square spot of the street and is directly associated with the value of its respective cell in the Attraction-Matrix.

To establish the route basis, it is necessary to reconstruct the weights of the underlying undirected graph, where the vertex set represents the street spots and the edge set reflects their adjacency according to the 7-neighborhood, while incorporating the influence of the defined attraction points.

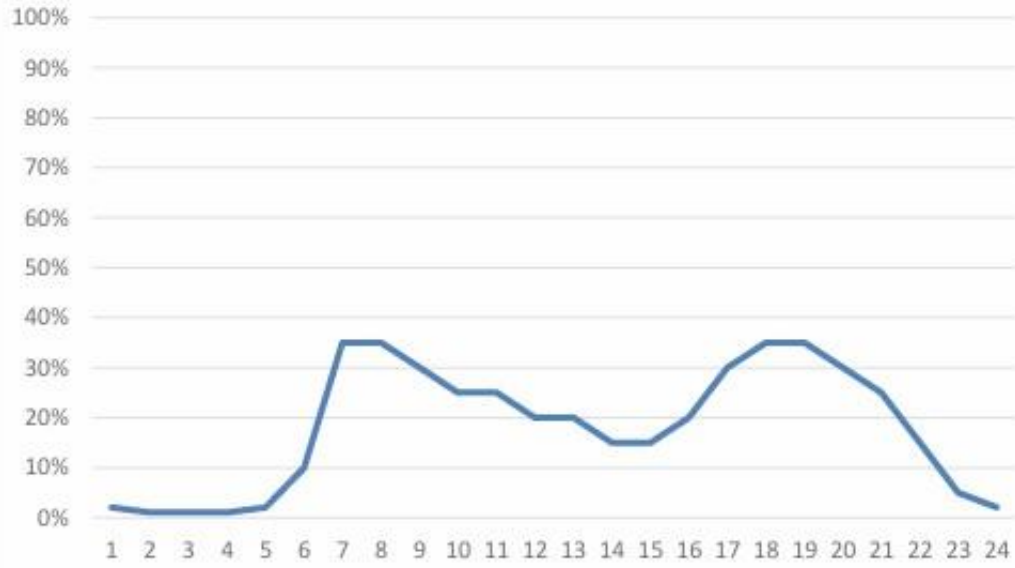


Figure 5. Citizen Mobility in Hourly Congestion

In figure 5 we see the temporal distribution of traffic intensity across a 24-hour period. The curve reflects the percentage of citizens moving through the city at each hour, with distinct peaks corresponding to the morning and evening rush hours. This distribution is integrated into the framework to ensure that route construction and mobility dynamics realistically follow daily traffic patterns, thereby capturing the natural variability of urban movement and its effect on epidemic spread.

The new weights for an edge $e_{i,j}$ connecting the vertex v_i with the vertex v_j are calculated by the following formula:

$$W(e_{i,j}) = 10 - L_i + 10 - L_j + 1, (1)$$

where L_i and L_j are the attraction matrix cell's values (i.e., $[0 - 10]$), casted on the vertices that connected by the corresponding edge. For example, let that we want to calculate the weight of the edge, e_1 . The procedure begins by finding the IDs of the nodes that e_1 connects. In our example let us assume that the vertex with $ID = (323,50)$ and the vertex with $ID = (324,51)$ are connected with the edge e_1 . The procedure continues by searching the values in the corresponding cells of the Attraction Matrix (i.e., in this example L_i is the value of the cell with coordinates $i = 323$ and $j = 50$ and L_j is the value of the cell with coordinates $i = 324$ and $j = 51$). It is crucial to note that, at the end of the procedure, all the weights of the graph will be constructed and according to our formula their corresponding values fluctuate from 1 to 21.

Considering two hot spots whose values are 10, using our formula to calculate the weight of the corresponding edge the output will be 1. Additionally, considering two cold spots whose values are 0 our formula will calculate the weight of the corresponding edge as 21. To conclude, in this section, our model reconstructs the edges of the graph considering the attraction points of the image and the calculation formula (the user has already set the attraction points in the initial image), whose weights ranges from 1 to 21 (i.e., the lower the weight of the edge, the hotter the corresponding spots.) in order to construct the basis for the routes of the city.

2.6 Mobility Model

The proposed model is designed to realistically simulate the spread of an airborne virus. The movement patterns of citizens, including the frequency of their outings and the specific routes they choose, are critical factors that influence the dynamic population density within the city's network over time. These variations in density directly affect the rate at which the virus propagates. This section presents a detailed description of the citizens' mobility schedules and the algorithms used to determine their individual paths.

2.6.1 Scheduling the Mobility of Citizens

Considering the table of Figure 5, which represents the percentage of the citizens moving in a city over specific time periods, our model schedules the mobility of citizens, securing a smooth and realistic simulation of citizens' movements (i.e., routes and paths). Considering that rush hours differ from city to city, our model is flexible enough to support tables with any hour peak, according to user demands. The first and the second column of Table 1 represent the time of the day and the corresponding simulation step, while the third column imprints the percentage of the moving citizens into the town. More precisely, every single citizen has a chance to start a walk at any time or minute during the day from the start point of the city to a random destination. The probability for each citizen to start moving into the city increases or decreases according to the percentage of the population permitted regarding the type of the hour (i.e., rush hour or not), based on Table 1.

2.6.2 Computing Shortest Paths

In this section we discuss the procedure that describes the assignment of the start and the destination points of each citizen's path (i.e., the routes that model the citizens' mobility), how points of interest affect those paths and possible alternative routes assignments as long as a path has already been set.

2.6.2.1 The effect of Hot-Spots on Citizens Routes

As we mentioned previously, we have constructed a graph from our initial image whose vertices represent square spots of the city routes and weighted edges represent the connections between them. The weight of each edge depicts the characteristic of the corresponding vertices (i.e., a very low weighted edge links two vertices, that possibly correspond to two neighboring hot spots in the initial image). Given a start and a destination point, our goal for each citizen, is to choose the less weighted (i.e., shortest) path among all possible paths that connect the start-destination points. Given a start and a destination point, the shortest, and hence the less weighted, paths in our graph correspond to the shortest of the city. Additionally, it is of major importance to refer that this approach corresponds to the paths that utilize the edges among the spots with greater values, or, in other words these paths contain the edges that are included in most of the paths. To achieve this, we use Dijkstra's shortest path algorithm for every citizen who decides to move from a start to destination point of the graph, ensuring hence that his path includes the less weighted edges on its sum.

2.6.2.2 Stochastic Approach

The procedure of the assignment of start and destination points in a walk for each citizen is a very crucial task, and hence of major importance for the spread of an epidemic. As long as the points of interest have been set and the weighted edges of the graph have been reconstructed, our system assigns to each citizen a static start point that remains unchanged, e.g., the residence of a citizen. The start point of a citizen is a point on the map that he starts and ends most of his routes representing somehow the location of his home.

When the home location of each citizen has been set, initially none of the citizens are moving across the roads of the city. Then, iteratively over specific periods of time every citizen has a probability to make an action (i.e., defining a path between its source and a destination point and start the corresponding route through the city) depending on his corresponding state (i.e., “stay-home”, or “get-out”). At the beginning of a period, later we will discuss this term by the aspect of simulation) all the citizens are located at their residences (i.e., home locations), so each citizen computes probabilistically on whether to stay at home or start a route in the proximal time. To this point we ought to mention that in terms of simulation, the term “period” refers to the simulation of a day, while the term “proximal time” refers to the next hour.

Then, each citizen whose route is scheduled to start, a random timeslot (the timeslot ranges from 1 to 60 minutes from the current time or 1-500 simulation steps from the current step) is assigned into, to start its route. Additionally, it is worth noting that a random destination point of the city’s map is assigned in order to construct the route of a citizen that is about to move, when the citizen is about to get out w.r.t the time defined by the timeslot assigned into him. Since destination point has been set, we deploy Dijkstra’s shortest path algorithm in order to calculate the shortest path of the graph and navigate the corresponding citizen from start to destination point via shortest and less weighted path. This procedure described some of the basic actions of people that our system supports through the mobility model.

2.6.2.3 Multiple Tasks in a Row

As we mentioned previously, our proposed model decides whether a citizen starts a route or not, within a proximal time w.r.t the probability defined for the corresponding time intervals depicted above. Focusing on possible actions that may occur during the route of a citizen, when it is time for a citizen to start his route, our system calculates the shortest path and the corresponding citizen placed into the Graph’s vertex that represents his start point of the city. At this point, in every simulation step the citizen moves from his current vertex of the Graph to one of its neighbors, more specifically, to the next vertex that Dijkstra algorithm undermines for the corresponding path. However, during his navigation on the streets, there is a chance to meet a known contact who also navigates to his destination point into the Graph. In this case, both citizens will stay at the same vertex for 0 to 5 time slots and then they will continue their navigation to their corresponding destination points. In our model to construct such a contact-network we construct a random graph using the Caveman model, depicting the friendships among the citizens. This procedure reflects a real-life scenario in which two people that know each other randomly meet in a city’s street and stand-by for at least a handshake, or a short talk. As long as the citizen reaches his destination point of the city, remains stable to the corresponding vertex of the graph for a specific period of time and then decides one of the following actions, i.e., either to move to a new destination point of the city using the procedure we mention before, or set his destination point as his home location, i.e., the vertex of the graph that he started his route. With this approach, our model is able to simulate real-life scenarios where during a random walk in a city, a

citizen may meet a friendly contact and pause his navigation for a short amount of time. In addition, our simulation considers the probability that a citizen's walk may include multiple destination points before he returns to his home location, as is the case in a real-life scenario.

2.7 Propagation Model

This section presents the proposed approach for modeling the spread of an airborne virus, structured around the key characteristics associated with its transmission behavior. The discussion includes the underlying epidemic model, the range of possible actions taken by individuals that may contribute to the spread of the virus, the progression of viral load within an infected host, and the internal evolution of the virus in relation to symptom development and the corresponding increase in host infectivity.

2.7.1 Underlying Epidemic Model

Our system incorporates the SIII epidemic model where each citizen can be one of the states: *Susceptible*, *Infected*, *Infectious*, *Immune*, implemented as follows:

Susceptible state, refers to individuals who never hosted the virus before and they are susceptible to it,

Infected state, refers to individuals who recently got infected by the corresponding virus and are not yet contagious,

Infectious state refers to individuals who are infected and contagious and

Immune state refers to individuals that are not characterized neither as Infected nor as Infectious and hence do not transmit the corresponding virus.

Investigating the states that are critical for the contagion of the virus, in our model we study four possible actions that can be performed by any citizen in order to spread the virus. The possible actions that our model supports are breathing, coughing, sneezing and talking and each one, has a probability of being performed at any time and its corresponding percentage of viral load to propagate is analogous to the evolution of the viral load inside the infected host. The probability of each action differs according to each citizen's state, his corresponding viral load and the corresponding virus inner characteristics. For example, an infectious citizen has got an increased probability of coughing than a citizen whose state is susceptible. In addition, the percentage of the propagated viral load from an action is also relevant to state and the corresponding viral load. For example, we assume three citizens whose states are susceptible, infected and infectious respectively. As we mentioned before, the citizens in the infectious state have an increased probability of coughing at a given time than the susceptible or even an infected one who is in the early states (i.e., infected one or two days ago, having low viral load).

In a scenario where three citizens perform the same action, such as coughing, the percentage of viral load released by an infectious individual is higher compared to that released by susceptible or infected (but not yet infectious) individuals. Additionally, both the probability of performing a given action and the amount of viral load spread by citizens in the same epidemiological state vary according to the rate of viral load evolution. The methodology for calculating these values will be discussed in the following paragraphs. In Tables 1 and 2 we provide estimation for the computation of the probability of an individual to perform an action that transmits infected droplets that carry the airborne virus, and additionally we cite an estimation on how this probability ranges across the evolution of the viral load inside an infected host, and the corresponding percentage of infected droplets. In Table 1 we present the probability of

an individual to perform one of the sets of basic actions that transmit airborne viruses through the emitted droplets, namely breathing, coughing, sneezing or talking, alongside with the corresponding volume of emitted droplets per action $VoED$. In Table 2 we present the evolution of the viral load (vl) inside an infected individual (see, column 1), the corresponding probabilities of performing an action that emits droplets (see, columns 2- 4), and finally, concerning the evolution of the viral load inside the host individual, the corresponding percentage of infected droplets $Perc(ID)$ emitted, that are independent of the performed action (see, column 6).

	Breath	Cough	Sneeze	Talk
Probability of performing an action	0.8	0.1	0.05	0.05
Volume in of Emitted Droplets ($VoED$)	$0.2 m^3$	$2 m^3$	$3 m^3$	$0.5 m^3$

Table 1. Probabilities of actions and corresponding volume of emitted droplets per healthy individual.

Percentage of Viral Load (vl)	Probability of Breathing	Probability of Coughing	Probability of Sneezing	Probability of Talking	Percentage of Infected Droplets $Perc(ID)$
$vl = 0$	0.8	0.1	0.05	0.05	0.1
$0 < vl \leq 10$	0.8	0.1	0.05	0.05	0.15
$10 < vl \leq 20$	0.8	0.1	0.05	0.05	0.2
$20 < vl \leq 30$	0.8	0.1	0.05	0.05	0.2
$30 < vl \leq 40$	0.8	0.1	0.05	0.05	0.2
$40 < vl \leq 50$	0.75	0.15	0.05	0.05	0.3
$50 < vl \leq 60$	0.70	0.20	0.05	0.05	0.4
$60 < vl \leq 70$	0.65	0.25	0.05	0.05	0.5
$70 < vl \leq 80$	0.60	0.30	0.05	0.05	0.6
$80 < vl \leq 90$	0.55	0.35	0.05	0.05	0.7
$90 < vl \leq 100$	0.50	0.40	0.05	0.05	0.8

Table 2. Probabilities of actions and corresponding percentage of emitted droplets per individual across the evolution of its viral load.

2.7.2 Stochastic Viral Load Evolution

In this section, we examine how each citizen's immune system responds to the airborne virus from the moment of infection, with a particular focus on the evolution of the viral load. In real-life scenarios, the same virus can affect individuals in significantly different ways. For instance, the progression of viral load in an individual with chronic diseases tends to be more aggressive compared to that of a young and healthy individual. To incorporate this important real-world variability into our model, each citizen is assigned a unique graph representing the evolution of viral load from the point of infection. This Viral Load Evolution Graph encodes information such as the rate at which the viral load increases within the host, the onset day of symptoms, and the distribution of viral load over the course of the individual's infectious period. These graphs are not uniform; they have been developed to reflect variations in viral behavior as observed by scientists in different patients. Through this approach, the model captures the individualized

progression of the virus in each citizen, while preserving the fundamental characteristics of the virus itself.

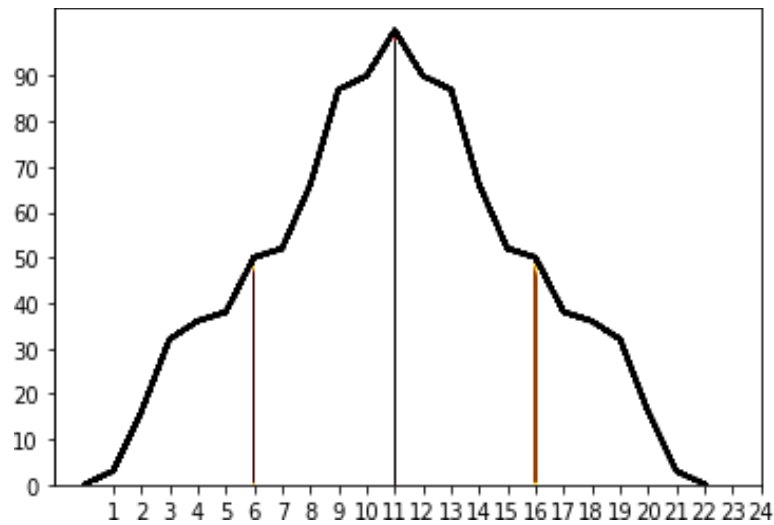


Figure 6. Viral Load Evolution Inside a Host.

A more detailed report on the construction of viral load evolution Graph will be discussed in paragraph 2.7.4.

2.7.3 Infection of Proximal Population

In this section, we focus on the procedure that describes the transmission of the air-born virus from an infectious to a susceptible citizen. Air-borne viruses spread through the air via respiratory droplets which produce after a citizen's action (i.e., talk, breathe, cough, sneeze). The quantity of produced droplets is proportional to the action the citizen makes, while the radius of each droplet can vary, depending on the intensity of the corresponding action. For example, the quantity of the produced droplets after a sneeze are up to 40000 whereas an action of cough may produce up to 3000. For further understanding, considering the increased spread of the droplets over distance, the droplets transition in the air can be described as a triangular cloud whose one corner located in the mouth, where the droplets are released. To make our model more realistic we consider the orientation of each citizen into the graph and develop his individual triangular cloud (see, Figure 8) which can transmit the virus inside its borders.

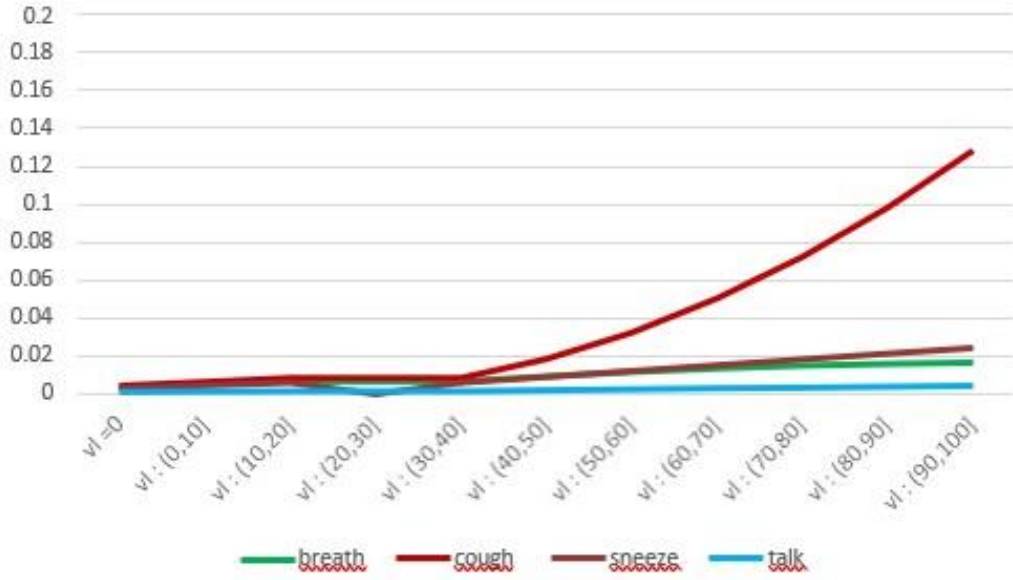


Figure 7. Probability of infection by a nearby host at distance 2m according to Table 1, Table 2, and Equation 1.

In this figure, we show the infection probabilities associated with different respiratory actions breathing, coughing, sneezing, and talking, under varying levels of viral load. The results highlight that coughing and especially sneezing produce significantly higher probabilities of transmission, as the larger volume and greater intensity of expelled droplets increase the risk for nearby susceptible individuals. In contrast, breathing and talking exhibit much lower probabilities, though they still contribute to transmission risk in crowded or prolonged interactions. This differentiation underscores the importance of modeling specific exhalation behaviors in capturing realistic airborne virus propagation.

To simulate the transition of the virus and the infection of proximal population we need to consider the distance between them, the viral load and the kind of exhalation (i.e., breath, talk, cough, sneeze). We develop a formula to calculate the probability of infection considering those three factors. The probability of an individual X to get infected $P(X \rightarrow \text{Infected})$ decreases as long as distance from nearby infected individual Y increases, while respectively increases as long as viral load and the quantity of droplets produced are increased.

The formula to compute the probability of an individual X to get infected by the action of a nearby individual Y that has a specific viral load and performs a specific action is computed as follows:

$$P(X \rightarrow \text{Infected}) = \frac{\text{Perc}(ID) \times \text{Perc}(VoED)}{1 + \text{Eucl}(X,Y)^2}, \quad (2)$$

where $\text{Perc}(ID)$ corresponds to the percentage of infected droplets a nearby infected individual emits by an action, $\text{Perc}(VoED)$ corresponds to the volume V of the emitted droplets a nearby infected individual emits by an action, and the $\text{Eucl}(X,Y)^2$ corresponds to the square of their in-between Euclidean Distance. For example, we assume two infectious people with the same amount of viral load and same orientation, abstain 5 meters from two susceptible citizens respectively. The first infectious citizen

sneezes while the second one coughs. The probability that the first infectious citizen will transmit the virus to the corresponding susceptible citizen is higher compared to the second one, considering the more quantity of respiratory droplets produced by sneezing over coughing while the distance, orientation and viral load remain the same.

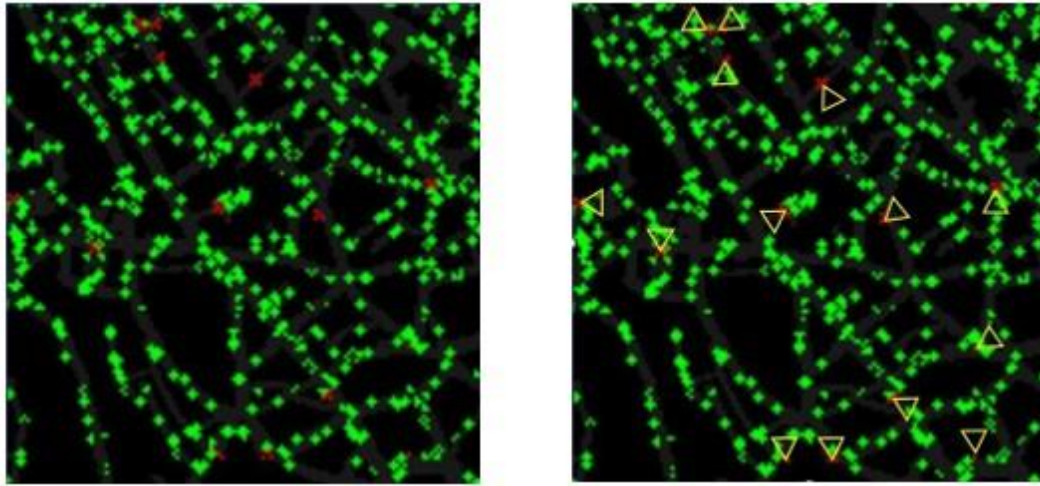


Figure 8. Distribution of Individuals (left) and Triangular cloud transmission of airborne virus (right).

This figure illustrates the epidemic progression in the simulated city. Susceptible individuals are shown in green, infected ones in red, while triangular markers indicate new exposures and the formation of infection clusters over time.

2.7.4 After Infection Insights

If a susceptible citizen gets infected, his viral load Graph (see Figure 2.8) depicts his viral load evolution during the days which determines the incubation period, his first day of symptoms, the day that his viral load reaches the peak and his cure day. According to recent scientific studies, it is estimated that an ailing is going to reach 50% of his viral load and show his first symptoms 3-6 days after his infection. From the day of symptoms there is a margin of 3-6 days for the viral load to increase and reach its peak. In our model, we consider that the decrease ratio of the viral load curve equals the corresponding increase ratio. As long as the percentage of the viral load of the corresponding infectious citizen increases, the probability of coughing and the percentage of the infected respiratory droplets increase analogously. As a result, the citizens become even more contagious and the probability of spreading the virus increases dramatically. We combine the information from Table 1, Table 2, and utilizing Equation 2 we represent the probability of infection by a nearby infectious individual that has distance 2m from a susceptible citizen.

Section 3. The Proposed Model

In the previous section, we analyzed the baseline simulation model. Building upon that foundation, this section presents the enhanced integrated simulation model aimed at supporting pandemic prevention and mitigation. The new version introduces multiple components that explicitly combine epidemiological modeling with computational mechanisms for contact tracing and area risk evaluation. Specifically, we implement functionalities for dynamic contact tracing with multi-level infection status propagation, monitoring citizen compliance through predefined obedience scenarios and localized risk assessment in critical zones, computed in real time using the mobility traces of individuals. In addition, we extend the navigation layer of the model to support adaptive route selection, where citizens are directed toward alternative paths whenever their current location is deemed high-risk according to the computed Area Risk Assessment.

3.1 Theoretical Background

Building upon the foundations of the SARiSsa framework [30], which integrates spatial modeling, population mobility, and a stochastic propagation model for simulating the spread of airborne pathogens, we adopted its modeling principles and extend them to further enhance epidemic risk evaluation. While SARiSsa focuses on simulating contagion through realistic urban maps, attraction points, and proximity-based interactions among individuals, in this work we propose an extension of the propagation model by incorporating an SARiSsa mechanism. The $A(r)$ metric, as introduced in the literature, quantifies the relative probability of exposure in localized regions by considering factors such as proximity to infected individuals, contacts of contacts, and spatiotemporal co-presence within a defined radius. By embedding this risk assessment procedure into the propagation model, we aim to capture not only direct transmissions but also the contextual risk associated with high-congestion areas or critical urban hotspots. This addition allows us to evaluate in greater detail how the dynamic interaction of individuals with their surrounding environment contributes to the likelihood of viral spread, thereby improving the overall predictive power of the framework and supporting more precise intervention strategies.

3.2 Augmenting the contact tracing procedure

Contact tracing constitutes one of the most critical procedures in epidemic mitigation, as it enables the identification and isolation of individuals who may have been exposed to infection before they further contribute to viral transmission. Traditionally, this process relies on retrospective identification of close contacts, often with delays that reduce its

effectiveness. Modern approaches, however, emphasize the integration of computational tools and mobile technologies to automate and accelerate this task. In our framework, this functionality is implemented through the updated contact tracing procedure, which provides a structured algorithmic mechanism for detecting and validating contact events within the simulated population. The function maintains a rolling record of spatiotemporal encounters between pairs of individuals and enforces a threshold condition that requires some consecutive steps of close proximity before an interaction is formally classified as a contact. This temporal constraint ensures that incidental or short-lived encounters do not inflate the tracing process, while still capturing the epidemiological intuition that sustain closeness is a significant factor in transmission risk.

Once the condition is satisfied, the function evaluates the health states of both individuals involved and applies a series of logical rules to update their epidemiological classification. If the first individual is infectious or infected, the contact is immediately marked, and the status of the partner individual is updated accordingly. Conversely, if the second individual is infected or infectious, the focal individual is upgraded to the “contact of infected” category. The algorithm also accounts for cases of recursive exposure: if the partner individual is already recorded as a “contact of infected,” then the focal individual is promoted to the “contact of contact of infected” class, thus broadening the tracing scope beyond direct interactions. To avoid overcounting and to preserve only recent encounters, the function continuously removes the oldest entry from its sliding window after each evaluation. Taken together, the contact control routine embeds into the simulation the essential dual role of contact tracing; first, as a preventive tool that captures both direct and indirect transmission chains in real time, and second, as a computational mechanism that enriches the accuracy of epidemic modeling. By encoding duration, proximity, and relational state updates in a single function, the model reproduces the dynamics of digital tracing systems while also providing a flexible basis for further extensions such as area risk assessment and adaptive interventions.

3.3 Area Risk Validation in Critical Zones

Assessing the risk of specific areas is a crucial step in understanding how epidemics spread across heterogeneous urban environments. Certain zones, such as transit hubs, crowded squares, or popular gathering points, can concentrate exposure and accelerate transmission. To capture these dynamics, the enhanced model incorporates an area-based risk assessment module that quantifies the probability of infection within localized neighborhoods. The area risk assessment is operationalized through calculating the area risk validation, which systematically evaluates the risk level of each individual's surrounding environment.

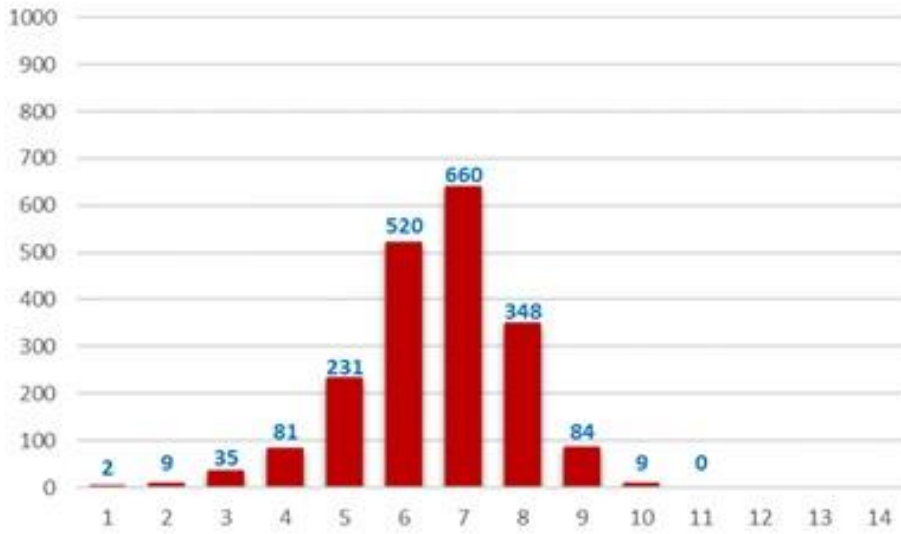


Figure 9. Averaged daily infection cases regarding the spread of an airborne pathogen taking to no countermeasure to control the spread.

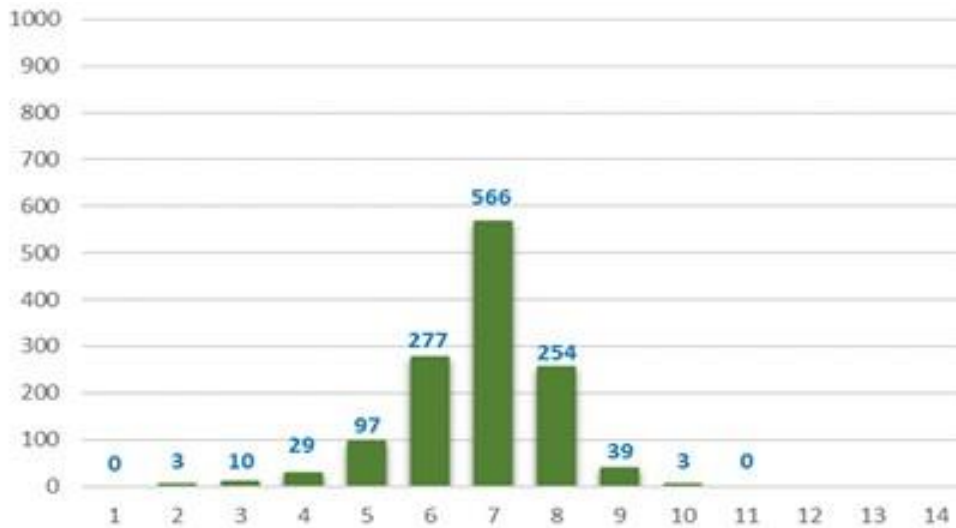


Figure 10. Averaged daily infection cases regarding the spread of an airborne pathogen taking by deploying the SARIssa smart application for contact tracing.

In Figure 9 and 10 we present Averaged daily infection cases regarding the spread of an airborne pathogen. The first represents the uncontrolled environment with no countermeasures, while the latter plot corresponds to the deployment of the SARIssa smart application for contact tracing, which reduces and delays the epidemic peak compared to the baseline scenario.

We begin by selecting a specific target area on the city map and enumerating the individuals currently located within their spatial bounds. To compute the area risk level, each agent is treated as a referencing point, and we scan a local radius around them. Specifically, for each such agent p at map coordinates (x_p, y_p) , we define a local neighborhood window, which is a fixed 10×10 rectangular region centered on (x_p, y_p) .

Operationally, this neighborhood is the set of positions (x_i, y_i) that satisfy the axis-aligned bounds:

$$x_{\min} < x_i < x_{\max} \quad \text{and} \quad y_{\min} < y_i < y_{\max}$$

where $x_{\min}$ and $y_{\min}$ are computed from (x_p, y_p) by extending 5m in each axis. This rectangular neighborhood acts as an efficient approximation of spatial radius queries, aligning with the grid-based structure of the simulated city map. All individuals whose coordinates fall inside this window are enumerated, and their infection-related status is retrieved from the global propagation model.

Inside this neighborhood, we scan all co-located individuals q , with coordinates (x_q, y_q) within the window, and determine their relationship to p and their health status. For each qualifying neighbor, the function increments one of four counters corresponding to the major epidemiological categories: infectious individuals, infected individuals, contacts of infected, and contacts of contacts of infected individuals. These counters serve a dual purpose: first, they allow the construction of a 1×4 binary status vector for each reference agent, indicating the presence or absence of at least one individual in each category. Secondly, they provide the raw numerical values needed for the risk scoring formula.

For every agent p inspected in this way, we the 1×4 binary feature vector is set as:

- [0] infectious: set to 1 if at least one person inside the 10×10 rectangle is currently infectious.
- [1] infected: set to 1 if at least one person inside the rectangle is currently infected.
- [2] contact of infected: set to 1 if at least one person inside the rectangle is a contact of infected.
- [3] contact of contact of infected: set to 1 if at least one person inside the rectangle is a contact of contact of infected.

It is important to emphasize that the counters are reset to zero at every iteration, ensuring that when scanning new neighborhoods, the vector reflects only the individuals relevant to that focal person. Thus, if multiple citizens fall under the same category, the corresponding entry is still marked as 1, but it does not persist between different focal individuals.

While computing the 1×4 vector, we also maintain counters to support the risk formula, one for each category (i.e. a counter for infectious, a counter for infected etc. for each individual). These counters are incremented for every qualifying neighbor q whose position satisfies the distance restrictions and whose health or relationship classification matches the definition above.

To formalize this process, we define four counters:

- number of infectious individuals in the radius
- number of infected individuals in the radius
- number of contacts of infected individuals
- number of contacts of contacts of infected individuals

Once all neighbors have been scanned, the algorithm computes the area risk assessment $A(r)$ using the formula:

$$A(r) = \frac{\sum_{i=1}^{n_A} w_A \cdot d_{A_i} + \sum_{i=1}^{n_B} w_B \cdot d_{B_i} + \sum_{i=1}^{n_C} w_C \cdot d_{C_i} + \sum_{i=1}^{n_D} w_D \cdot d_{D_i}}{\sum_{p=1}^N w_A \cdot \bar{D}}$$

where n_A , n_B , n_C , and n_D correspond to the cardinalities of the sets of the four categories of individuals, i.e., infected individuals, contacts of infected individuals, contacts of contacts, and random individuals, respectively, w_A , w_B , w_C , and w_D correspond to the weights of each category, i.e., infected, contacts of contacts and random individuals, d_{X_i} corresponds to the distance between the referencing point and the i -th individual of category X , let X_i , N corresponds to the total number of individuals (i.e., number of devices scanned nearby), and $\bar{D}_A$ corresponds to the average distance between the nearby detected individuals and the reference point, if the nearby detected individuals were all infected.

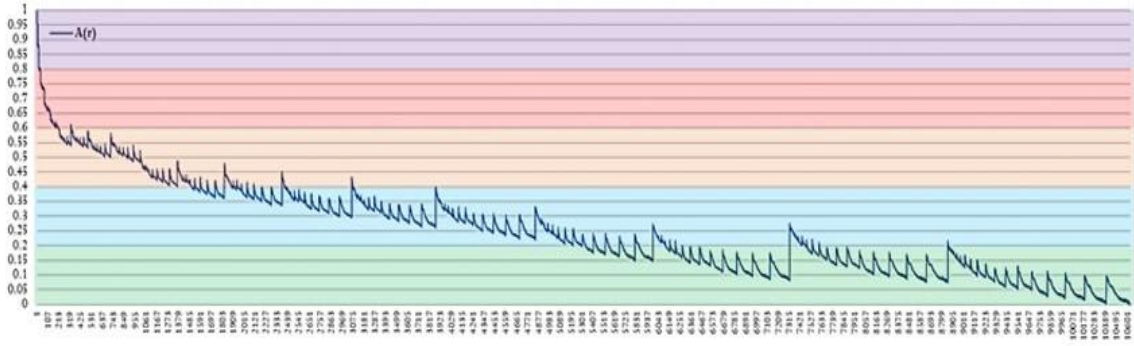


Figure 11: Study on the behavior of the risk assessment $A(r)$ for different combinations of individual categories.

The curve of Figure 11 illustrates how localized risk gradually decreases over time as agents move away from high-congestion areas, while periodic spikes indicate temporary increases in exposure risk when individuals re-enter critical zones. The color bands provide a visual segmentation of different risk levels, highlighting the transition from high- to low-risk environments throughout the simulation.

Through this mechanism, the Area Risk Assessment module extends the baseline propagation model by embedding localized environmental awareness into each agent's decision-making process. This ensures that epidemic spread is no longer modeled only as a result of direct contacts, but also as a consequence of the broader spatial and behavioral context in which individuals interact. In this way, the model captures critical dynamics of urban epidemics, where crowding, mobility patterns, and heterogeneous compliance jointly shape transmission outcomes.

3.3.1 Incorporating Behavioral Compliance

An additional component of the proposed model is the obedience mechanism, which introduces behavioral heterogeneity into the simulated population. At the beginning of each simulation run, a predefined percentage of agents is randomly selected to represent the obedient subset. These individuals comply with the system's alerts generated by the Area Risk Assessment module: when they are detected within a critical zone, they either reroute to a safer path or terminate their trip and return to their origin.

This random selection is repeated for every run, ensuring that obedience is treated as a stochastic factor rather than a fixed attribute of specific agents. By incorporating this mechanism, the model captures the variability of real-world compliance and allows us to systematically investigate how different adherence levels influence the effectiveness of area-based risk mitigation.

Section 4. An Algorithmic Framework for Pandemic Prevention

4.1 Design Principles

In this section, we make a representation of the construction of our model in order to simulate the spread of an air born virus in a completely uncontrolled environment. The proper functionality of our framework is based on the combination of three different models, the city's spatial model, the mobility model that simulates the movements of the citizens inside the city based on the topology of the routes and the propagation model that simulates the spread of the virus among the citizens.

The city's spatial model is responsible for the transformation of the city, which is represented in the initial Google-Map image, into a Graph, considering the corresponding hot spots. Throughout our experiment series, we use a Google-Map image representing an area of a city, where we study the spread of the virus. It is important to note that the more detailed and distinct representation of the roads the image shows, the more precise and realistic the construction of the spatial model will be. The system transforms the initial Google-map image into an undirected Graph named Gmap, whose vertex set corresponds to a square spot of the street (i.e., pixel of the image), while an edge belongs to the edge set of the Graph if the corresponding end points are neighboring square spots., as we previously described in Section 2.5. As long as the Gmap and the points of interests have been constructed, we construct the mobility model to simulate the spread of a virus in an uncontrolled environment. We schedule the movements (i.e., routes) of the citizens as described in Section 2.6. Assuming the worst-case scenario, where people are not aware of the spread of an air-borne virus, the scheduled routes of citizens are irrelevant to state. For example, an infectious citizen will move through the city and perform his scheduled routes until the end of the experiment, even if he has fever or is symptomatic. The propagation model governs how the virus is transmitted during interactions. At each discrete simulation step, the model checks whether two individuals are located within the same or neighboring vertices of the graph, and if so, it evaluates the possibility of transmission. Transmission events are probabilistic, depending on the health status of the interacting pair (susceptible, infectious, infected, immune) and the duration of their co-location, which is tracked by the contact detection procedure. Sustained proximity across multiple steps increases the likelihood that exposure results in infection, reflecting the epidemiological principle that both distance and time contribute to risk. Once transmission occurs, the newly exposed individual's status is updated and propagated through subsequent steps of the simulation. Importantly, the propagation model remains invariant for a given airborne

pathogen, maintaining its transmission logic across all experiments, while only the mobility layer changes to test different preventive strategies. This design ensures consistency in disease dynamics while allowing us to isolate and study the effects of behavioral or policy interventions.

4.2 Modeling Spread in an Uncontrolled Environment

To model the spread of a virus in an uncontrolled environment, we create the corresponding weighted undirected Graph from the initial image including hot spots and model the citizens, to navigate into it following random walks. The frequency of each citizen's routes is developed according to the corresponding mobility table, ranging from 3 to 4 routes per day and this number remains unchanged throughout the whole extent of the experiment independently the state of the corresponding citizen.

During a simulation day, as long as citizens move through the city, they may take one of the actions which are described in Section 2.7, spreading hence the air-borne virus to the susceptible citizens. The virus propagation in each infected citizen is described from his viral load evolution chart (see, section 2.7.2) which defines how contagious the corresponding citizen is. The experiment terminates as long as and 95% of the population has been infected with the air-borne virus and we define it as a pandemic.

4.3 The Effect of Density Reduction Measures on Pandemic Prevention

Conductive to reduce the spread of the airborne virus or even prevent a possible pandemic in the initial city we implement in our simulation model and more specifically in citizen's mobility model, the deployment of Density Reduction Measures. The construction of the spatial model and the virus propagation model to examine the influence of density reduction remain the same since we examine the spread of the same virus in the same city and population. The aim of this implementation is the reduction of the spread of the airborne virus decreasing the density of the citizens residing and moving through the city.

To achieve this, we allow, at most, 50% of the people of the initial mobility table to move through the city during the extent of a day. For example, at 12 o'clock in the corresponding mobility table of the first experiment approximately 20% of the citizen's population move inside the city performing their scheduled routes. Applying Density Reduction Measures, if a citizen has scheduled a route at 12 o'clock and 10% of the population are moving into the graph, the corresponding citizen will cancel his route ensuring that the percentage of the citizens that are moving inside the city at that specific time does not exceed 10% of the initial population.

4.4 The Impact of Area Risk Assessment on Pandemic Mitigation

In order to reduce the spread of the airborne virus or even prevent a potential pandemic in the initial city, we extend our simulation framework with the Area Risk Assessment feature, implemented within the propagation model. The construction of the spatial and mobility models remains unchanged, since we continue to study the same virus within

the same urban environment and under the same population dynamics. The objective of this feature is to mitigate viral spread by evaluating the riskiness of each citizen's immediate surroundings and, when necessary, guiding them to avoid traversing highly infected traffic areas.

The process is based on a localized scanning procedure around every individual. For each citizen on the city map, the system defines a fixed-size rectangular window centered on their current position. All neighbors within this area are enumerated and classified according to their health status and relationship group, as defined in Section 3. Each focal individual is then assigned a four-dimensional vector, updated according to the presence of qualifying neighbors, while counters are reinitialized at every step to ensure that the vector reflects only the current state of the environment.

This formulation allows the model to dynamically compute the area risk score $A(r)$, representing the epidemiological intensity of the citizen's local neighborhood. Citizens then respond according to their obedience factor: highly obedient individuals follow the model's recommendation and actively change their path to avoid high-risk areas, while less obedient ones may ignore the warning and maintain their original trajectory. In practice, this means that when $A(r)$ exceeds a threshold, obedient agents are not only notified but are also redirected to alternative, safer destinations within the city.

Through this process, the Area Risk Assessment feature extends the propagation model by coupling local environmental awareness with behavioral adaptation. This not only enables more realistic simulations of viral transmission but also demonstrates how targeted, zone-level early-warning mechanisms can reduce contagion probability in densely populated areas and contribute to pandemic suppression.

4.5 Flatten the Curve through a Spread Aware Application

A widely adopted preventive measure to curb the transmission of airborne viruses is the isolation of infected individuals for a defined period, aiming both to facilitate recovery and to limit further spread across the urban area. Within the context of our simulation, we presume that each citizen enters quarantine two days after the onset of initial symptoms (i.e., when the individual's viral load curve reaches 50%). This quarantine period spans 10 days, during which the affected citizen is restricted from moving, interacting with others, transmitting the virus, or contracting it from others. Once this quarantine period concludes, the individual's status transitions from infectious to immune, allowing them to resume movement through the city as per their predefined trajectories. Building upon our prior methodology and incorporating fundamental principles of infection monitoring, we propose the use of a spread-aware mobile application. This application, operating on smart devices equipped with GPS and Bluetooth 4.0 or higher, is capable of tracking the spatial positions of users during their movements, thereby logging encounters to aid in infection tracing and potentially mitigate or postpone a pandemic scenario. Every citizen utilizes this application, which continuously monitors their daily interactions and logs them into a centralized data system. Recorded contact is defined as any individual within a 10-meter radius of the respective citizen. Additionally, the application records the duration of each contact with other individuals.

Within this framework, it is assumed that each citizen will be placed under quarantine two days following the emergence of symptoms, along with any individuals they were in contact with over the previous two days, provided those interactions lasted longer than 12 minutes. Each quarantine lasts for 10 days. During this period, the citizen's planned activities and movements are suspended, and they are restricted from traversing the

city or transmitting the virus to others. Upon completion of the quarantine, the citizen may resume movement throughout the city based on the mobility schedule, and their state will be updated accordingly: if the individual was susceptible before entering quarantine, their status remains unchanged; otherwise, they transition to an immune state. As stated in previous sections, the structure of the spatial model and the virus propagation mechanism remain unchanged, as the initial city layout, population, and characteristics of the airborne virus are considered fixed.

Section 5. Evaluation

In this section, we present the results obtained by running a series of Monte Carlo simulations based on our integrated framework and analyzing their impact on epidemic outcomes. The experiments evaluate the implemented countermeasures in terms of their effectiveness for pandemic prevention and are further compared to the performance of the 2D-Co-Contact algorithm proposed by Nikolopoulos et al. [47]. This comparative analysis highlights both the strengths and limitations of our approach, providing insight into the relative contribution of each strategy.

5.1 Air- Born Virus Spread in an Uncontrolled Environment

In this section we will discuss the results of our experimental simulation, after the application of the air-borne virus spread in an uncontrolled environment model (see section 4.2). The following tables represent the average results after 5 simulations of the spread in an uncontrolled environment model. We set 1 individual as infected and 1999 susceptible people moving according to the movement model into the city which represented at Figure 12. At first figure (Figure 12 (left)), the x-axis represents the number of days, and y-axis imprints the number of people that got infected the corresponding day. For example, on the 4th simulation day of our experiment, we observe from the corresponding bar plot that approximately 60 people got infected. In Second figure (Figure 12 (right)) the x-axis also represents the corresponding simulation day and y-axis the overall number of people being infectious. For example, there are about 278 people that their corresponding state is infectious from the start of the experiment until day 6. As we can see in both figures, our simulation experiment ends after the completion of 10 days. In the first image we see a gaussian distribution. We observe an increase of the daily infected people until day 7 and a sharp decrease over the days 8, 9 and 10. In the second image we observe a low increase over the 5 first days, an exponential increase of the infectious people until day 9, a logarithmic increase from day 9 to day 10 and a stable line over the next days as long as all people have been infected.

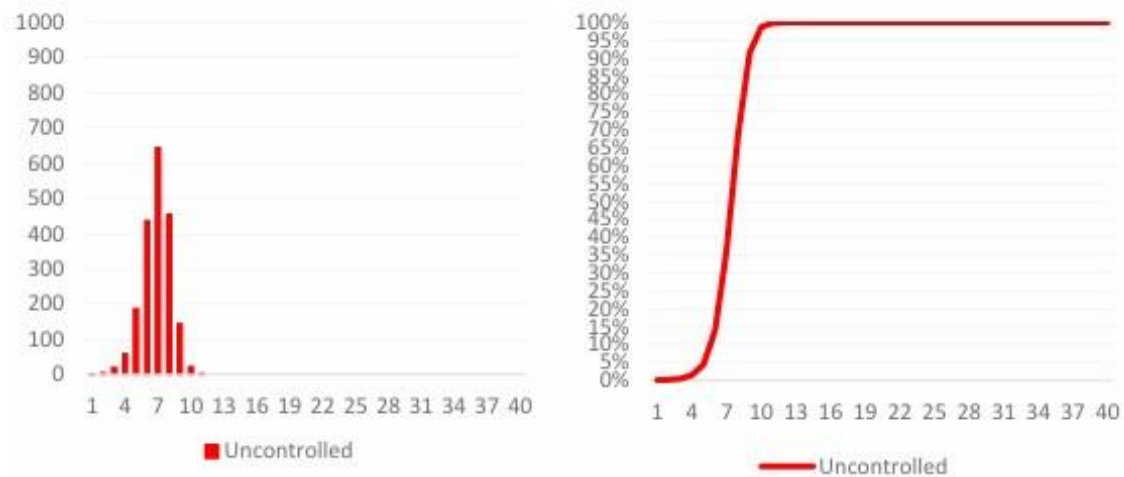


Figure 12. Pandemic Spread in an Uncontrolled Environment

These are the expected results as long as the spread of the virus evolves into an uncontrolled environment.

5.2 Pandemic Prevention Utilizing Algorithmic Counter Measures

From the evaluation of our initial experiment, we observe that an airborne virus with the characteristics described cannot be considered viable once the entire population of the city becomes infected. To mitigate or prevent such a pandemic scenario, it is necessary to design algorithmic countermeasures and systematically study their impact on the city and its population.

5.2.1 Density Reduction

In this paragraph we will examine the average results after 5 simulation experiments, the density reduction measure in the corresponding population and city. The Density reduction algorithmic and the corresponding constructions of the movement spatial and propagate model have been described in previous sections (i.e., see section 4.3). The first picture (figure 12 (left)) represents the people that got infected each day. The second picture (figure 12 (right)) represents the infectious people from the start of the experiment until the corresponding day, exactly like the corresponding pictures of section 5.1. At both pictures we observe that the experiment ends after 27 simulation days. First picture imprints, a Gaussian distribution with its peak on day 18. At second picture we observe a sigmoid plot which increasing exponentially from simulation day 13 to 20 and logarithmically increases from simulation day 20 to 25.

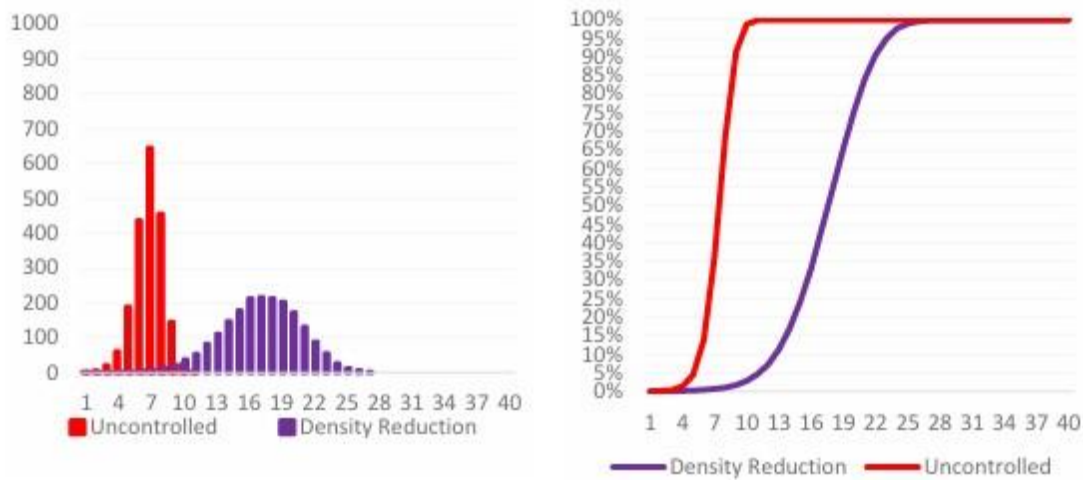


Figure 13. Pandemic Spread in an Uncontrolled Environment and in an Environment, where are applied the countermeasure of Density Reduction of population.

At the end of this experiment all the initial population got infected after 27 simulation days and delayed the pandemic. However, the implementation of the counter measure of density reduction in our simulation model was not enough to prevent a pandemic.

5.2.2 Spread Awareness

Another popular counter measure in order to prevent a possible pandemic is the Personal Quarantine of the infected people. In the terms of our experiment, we assume that every infectious person will be quarantined for 10 days, two days after their first symptoms. At the end of their quarantine the corresponding people have been cured, and they return to their daily routine walks with their corresponding state as immune. At the end of the simulations of Personal Quarantine counter measure, after observing figure 12 and figure 13 we cannot assume a significant difference after the application of this counter measure. The pandemic prevention failed and the daily infected and the infectious people are similar over the days compared with the results of the viral spread in an uncontrolled environment.

In the following graph, we observe two cases. The left graph (a) shows daily infection cases, where both uncontrolled and contact tracing scenarios exhibit a sharp epidemic peak within the first 10 days. The right graph (b) illustrates infectious population growth: while contact tracing slightly reduces and delays the peak, most of the population still becomes infected. Overall, the countermeasure proves insufficient to fully prevent the pandemic, showing limited effectiveness compared to the uncontrolled case.

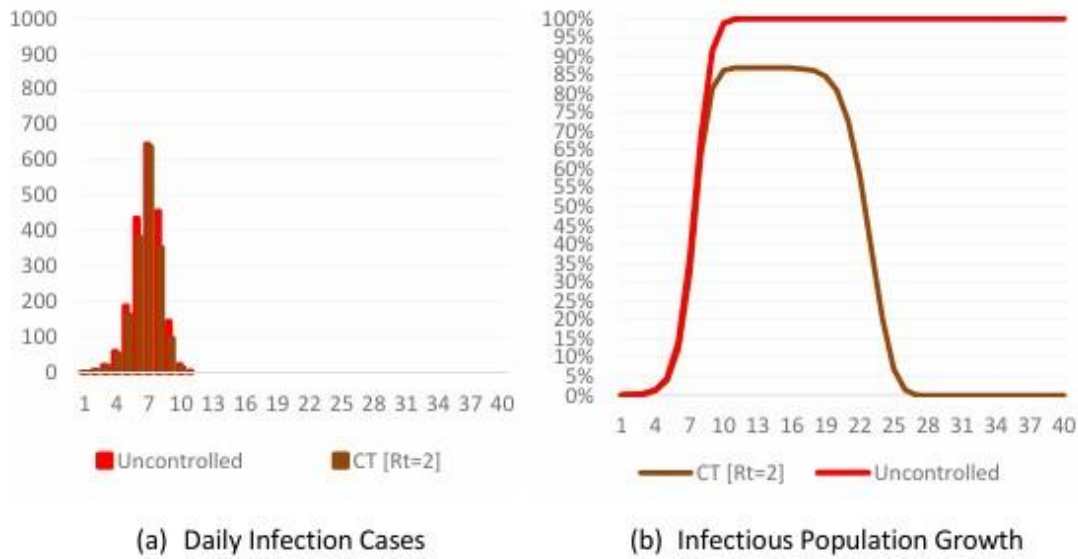


Figure 14. Pandemic Spread in an Uncontrolled Environment and in an Environment, where are applied the countermeasure of Spread Awareness with Contact Tracing.

5.2.3 Combination of Counter Measures

In this paragraph we will discuss the results of our experiment after the application of the combination of two counter measures to prevent the spread of the virus. The combined counter measure we use includes the Density reduction measure and the spread aware application which has been described in previous paragraphs (see section 5.3, section 5.5).

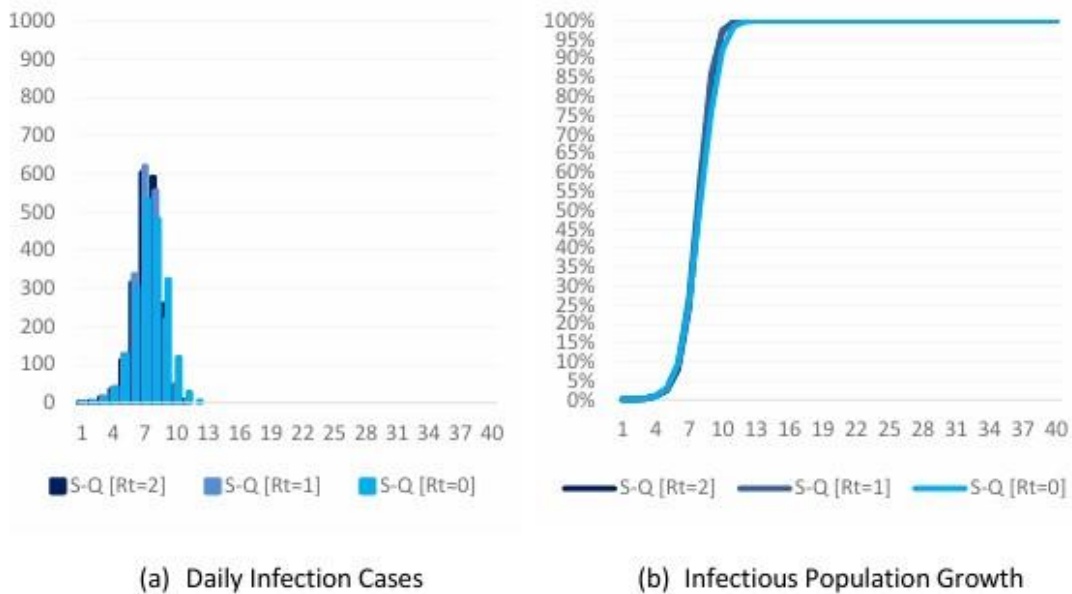


Figure 15. Pandemic Spread in an Uncontrolled Environment, where are applied the countermeasure of Spread Awareness without Contact Tracing for various Response Time values (i.e., $R_t=2$, $R_t=1$, and $R_t=0$).

Above, the left plot (a) shows daily infection cases, where the three scenarios behave almost identically, producing a sharp peak around days 6–8. The right plot (b) illustrates infectious population growth, reaching nearly 100% of the population in all cases. The variation of response time values (R_t) has no substantial effect, and the countermeasure fails to prevent or mitigate the pandemic.

More precisely, until day 10 the number of the infected population is negligible. From day 10 we observe a linear increase in the curve that reaches its top at day 25 with a ratio of 30% infectious citizens over the initial population. From day 25 until the 40th day, the curve decreases and reaches linearly 0 infectious citizens. At the end of the experiment the air borne virus has been removed since the state of the initial population has been transformed to immune during quarantine or remained susceptible, and the pandemic has been successfully prevented.

5.3 Area Risk Assessment for Pandemic Mitigation

For this set of experiments, we simulate the spread of the virus using one quarter of the total population, corresponding to 500 individuals, while simultaneously restricting the environment to one quarter of the city map. This proportional scaling ensures that the overall population density remains constant compared to the full-scale scenario. The experiment is conducted over a fixed period of 40 days, preserving comparability with previous experiments and allowing us to observe the dynamics of the outbreak. In doing so, the experiments preserve the same spatial and epidemiological dynamics while reducing computational complexity, letting us focus on evaluating the impact of the Area Risk Assessment feature under controlled and representative conditions. The results are derived from multiple Monte Carlo simulation runs and are averaged to ensure reliability and robustness. This experiment builds directly on the integrated spatial, mobility, and propagation models introduced in earlier sections, extending them with localized scanning and adaptive mobility responses based on the computed $A(r)$ values. The objective is to assess how this feature impacts epidemic dynamics by actively guiding citizens away from high-risk areas, thereby reducing exposure opportunities and potential transmission chains. Furthermore, the evaluation compares the outcomes of our implementation with the 2D-Co-Contact algorithm proposed by Nikolopoulos et al. [47], allowing us to highlight both the relative advantages and limitations of area-based risk assessment in relation to established tracing-oriented countermeasures.

5.3.1 Evaluation Metrics

To assess the effectiveness of the Area Risk Assessment feature, we employ standard epidemiological metrics that capture both the daily progression and the overall impact of the epidemic. Specifically, the model records the number of new daily infections, which allows us to observe how the spread evolves over time and to identify peaks and turning points in the outbreak. In addition, we calculate cumulative statistics, reflecting the total number of individuals who have been infected by the end of the simulation, and the basic reproduction number R_0 , which provides a measure of the average number of secondary infections caused by each infectious individual. These metrics were chosen because they jointly capture both short-term dynamics and long-term outcomes. To account for the inherent stochasticity of the simulation, each experiment is repeated three times using Monte Carlo sampling, and we report the average results across these runs to ensure robustness and reduce the influence of random fluctuations.

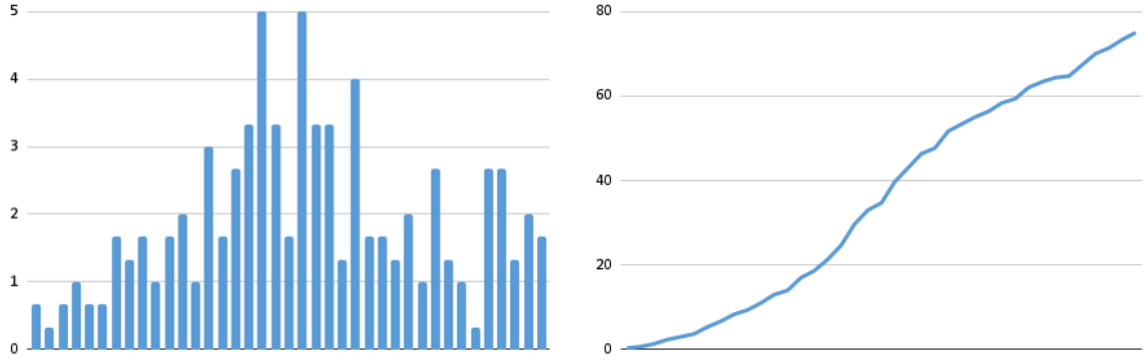


Figure 16. Daily new infections using the 2D-Co-Contact algorithm and cumulative infections graph.

In the baseline scenario, where the 2D-Co-Contact Algorithm is applied, the simulation results confirm the expected dynamics of an uncontrolled epidemic. Starting with a single infected individual, the number of new daily infections increases rapidly during the first week, forming a Gaussian-shaped curve with a clear peak before declining as the susceptible population is depleted. Infections grow steadily in an S-shaped pattern until the entire population becomes infected, while the reproduction number R_0 fluctuates during the early days before stabilizing near zero as transmission opportunities vanish. Overall, the baseline experiment demonstrates that without interventions, the airborne virus inevitably spreads throughout the population within the 40-day simulation horizon.

5.3.2 Risk Validation

In this paragraph, we extend the baseline simulation by incorporating the Area Risk Assessment feature together with the obedience mechanism. At the beginning of each simulation run, a fixed percentage of the total population is randomly selected to represent the group of obedient individuals. This randomization is repeated across runs, ensuring that obedience is not tied to specific agents but is instead treated as a stochastic factor, reflecting the variability of compliance in real populations. For example, when obedience is set to 30%, a new random subset of approximately 150 individuals (out of the 500 agents used in the experiment) is designated as obedient in that Monte Carlo iteration.

During the simulation, the system continuously evaluates the area risk validation $A(r)$ for each agent based on the density and infection status of nearby citizens within their scanning window. When an obedient individual is detected inside a critical zone, defined as an area where the computed $A(r)$ value exceeds the threshold of 0.6, the model enforces an adaptive behavioral response. Specifically, the agent may be rerouted to an alternative destination that bypasses the risky zone, or in cases where rerouting is not feasible, the individual terminates their trip and returns to the origin point. This adjustment directly reduces the probability of prolonged exposure in high-risk environments. Non-obedient individuals, by contrast, ignore these alerts entirely and continue along their scheduled mobility routes without modification.

By combining randomized assignment of obedience with dynamic risk-based interventions, the experiment captures both the heterogeneity of human compliance

and the spatial-temporal dynamics of epidemic risk. This design allows us to systematically study how different compliance levels influence the effectiveness of area risk validation, and to evaluate whether even partial adherence can significantly reduce the number of contacts in critical zones and delay or suppress epidemic peaks.

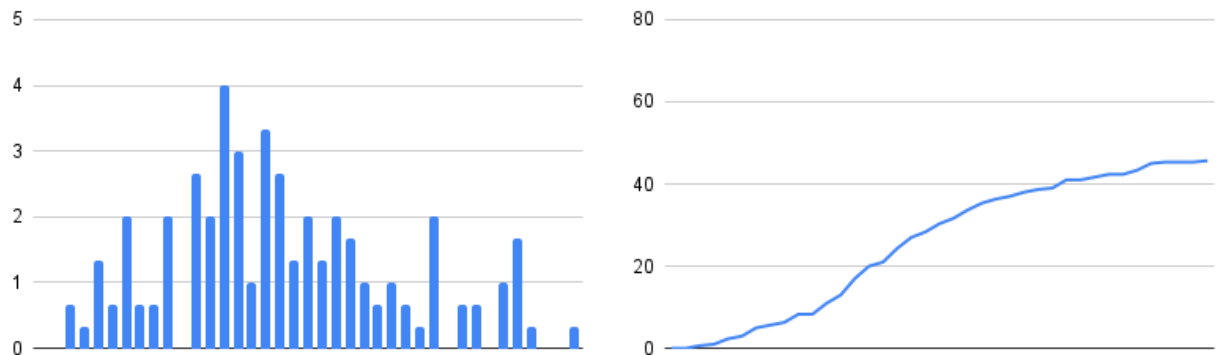


Figure 17. Daily new infections using the 2D-Co-Contact algorithm and the Area Risk Assessment with 30% obedient population and cumulative infections graph.

With 30% of the population being designated as obedient, the simulation results show a clear reduction in epidemic intensity compared to the uncontrolled baseline. Across the Monte Carlo runs, the daily new infections display a delayed and flattened peak, reflecting the effect of agents rerouting or returning upon encountering high-risk zones. The cumulative infection curves grow at a slower rate, demonstrating that partial compliance reduces the overall number of exposed individuals during the 40-day horizon. The average effective reproduction number R_0 also declines earlier in the simulation, indicating that obedience to area risk warnings contributes to breaking transmission chains.

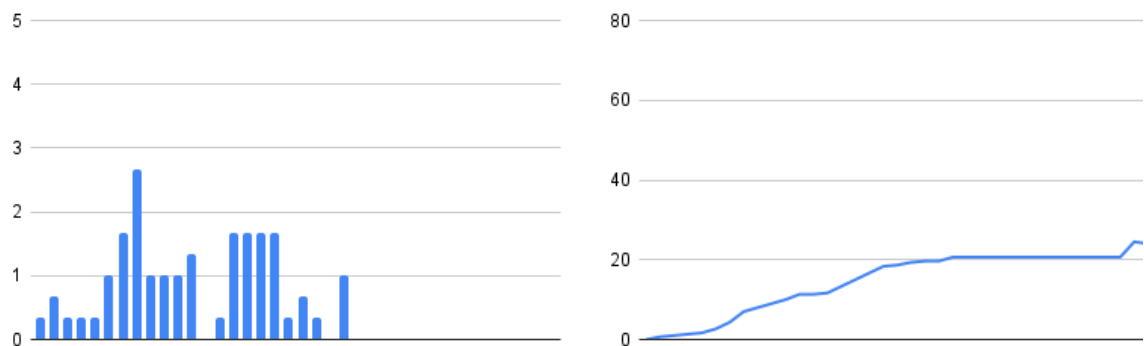


Figure 18. Daily new infections using the 2D-Co-Contact algorithm and the Area Risk Assessment with 50% obedient population and cumulative infections graph.

With 50% of the population set as obedient, the simulation shows a stronger suppression of the epidemic compared to both the baseline and the 30% case. Daily infections peak later and at lower levels, while cumulative infections rise more slowly, leaving fewer total cases after 40 days. The reproduction number R_0 also declines earlier, confirming that broader compliance with area risk alerts significantly limits transmission.

5.4 Discussion over Exhibited Results

In this section, we analyze the behavior of the airborne virus propagation in an environment without countermeasures and discuss the effectiveness of the implemented interventions. The evaluation includes a direct comparison between the outcomes of the baseline model and those produced by the proposed area risk calculation, highlighting the differences in epidemic dynamics under each scenario.

5.4.1 Behavioral Dynamics of Uncontrolled Spread

We now present the outcomes of the simulation experiments and analyze the epidemic curves following the application of the countermeasures described earlier, designed to prevent the onset of a pandemic. In the uncontrolled scenario, the virus spread aggressively, infecting the entire population by the tenth day. Similar results were observed when only infected citizens were quarantined, without extending the measure to their contacts, as this approach proved insufficient to limit transmission and halt the epidemic. When density reduction was introduced in the city, the spread of the virus was slowed, delaying the infection of the entire population by approximately twelve days.

Although the number of infected individuals per day decreased and the epidemic unfolded more gradually compared to the uncontrolled scenario, this measure alone was not capable of preventing a pandemic. More encouraging results were achieved through the spread awareness application, which combined the quarantine of infectious individuals with the isolation of their immediate contacts. During the first seven days, the growth curve of the infectious population resembled that of the uncontrolled environment, since no intervention had yet taken effect. From the day of activation, two days after symptom onset, the daily infections decreased sharply, and the cumulative number of cases stabilized. By significantly reducing the density of circulating individuals, this quarantine measure slowed down the spread of the virus and protected a portion of the susceptible population who were removed from exposure after being identified as contacts. This intervention provided sufficient time for infectious citizens to recover, develop immunity, and thereby prevented the complete saturation of the population.

Finally, the combined countermeasure, which integrated density reduction with the spread awareness application, produced the strongest results. The initial growth of infections was suppressed through density reduction, while the subsequent quarantine of infected individuals and their contacts kept infection levels consistently low throughout the duration of the experiment. Together, these interventions successfully limited viral transmission and prevented the development of a pandemic in the simulated city.

5.4.2 Evaluating the Effectiveness of the Fundamental Countermeasures

In this part we discuss the behavioral analysis of the pandemic over an uncontrolled environment and after the application of the critical countermeasures that are mentioned in the baseline framework. The spread in an uncontrolled environment is rapid and the pandemic after a few days over the initial city is unavoidable. The counter measure of the density reduction at first days seems to be efficient since the number of daily infected cases are at low levels. However, as the simulation days went by the cases increased until all the population got infected. The pandemic was delayed but not prevented. The first counter measure that successfully prevented the pandemic was the

use of the spread aware application which can track the infectious citizen's close contacts. Until the first day of the first quarantine the behavior of the pandemic was similar the behavior of the uncontrolled environment. After the day of the application of the quarantine the pandemic recedes. However, most of the population got infected even though this counter measure at last, was able to prevent the pandemic. The combined measure of spread aware application and the density reduction was the most efficient counter measure overall since the number of the daily infected cases remained at low levels and most of the population avoid the infection until the pandemic prevention and the end of the experiment.

Regarding the Grow-level in Figure 15 we observe an early increase from day 5 in the experiments of uncontrolled environment, the counter measure of self-quarantine and contact tracing. The experiment with the counter measure of response time increased its corresponding grow level approximately at day 11 since the density reduction delays its growth day compared to the counter measures that we mentioned before. The experiment with the combined counter measures (contact tracing and density reduction) performs better than the previous counter measures since shows a slight increase in its corresponding grow level on day 15.

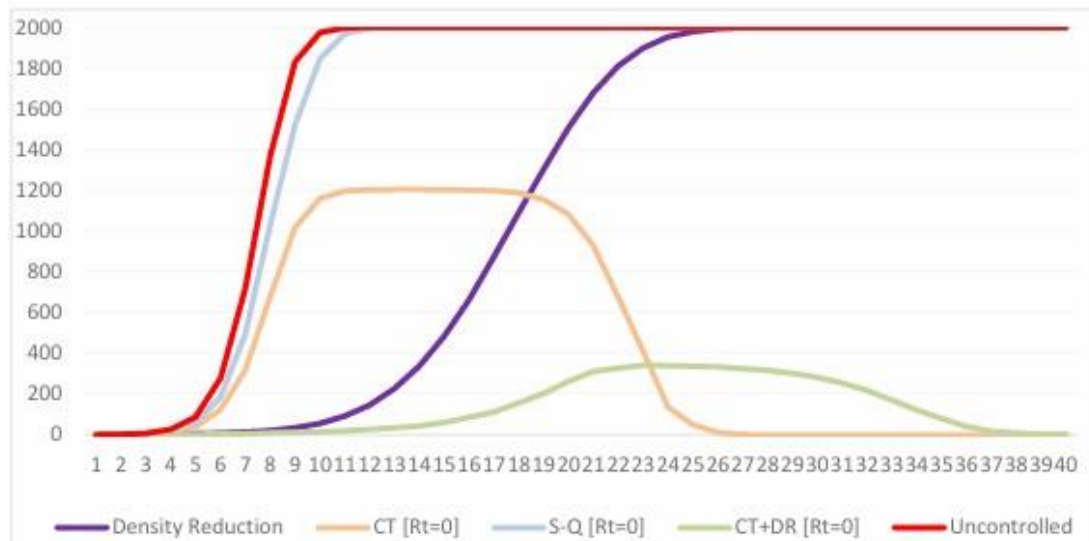


Figure 19. Comparative results among all the proposed countermeasures.

Concerning the peak of confirmed overall cases in Figure 19, we observe similar behavior among the experiments with the uncontrolled environment, self-quarantine, and contact tracing countermeasures, as their respective peaks occur around days 9, 11, and 12. On the other hand, the experiments with combined countermeasures and density reduction perform significantly better, reaching their peaks later, at approximately day 24, with about 1200 and 380 cases respectively.

Regarding the duration of high infection levels in Figure 19, the uncontrolled environment, density reduction, and self-quarantine scenarios perform the worst, as they all evolve into full-scale pandemics. The contact tracing countermeasure maintains high levels for about 8 days, whereas the combined countermeasure extends this duration to 11 days.

Finally, considering the overall duration of the epidemic, the uncontrolled environment and self-quarantine experiments again perform the worst, both reaching pandemic levels by day 11. The density reduction measure delays but does not prevent the pandemic, as by day 24 nearly all individuals are infected. In contrast, the combined

countermeasure prolongs the epidemic to 37 days while successfully preventing a full outbreak, whereas contact tracing prevents the pandemic earlier, at approximately day 26.

5.4.3 Assessment of the Effectiveness of the Proposed Approach

We analyzed the behavior of the pandemic under the baseline uncontrolled environment, the plain 2D-Co-Contact algorithm, and the enhanced approach where the 2D-Co-Contact mechanism is combined with the proposed Area Risk Assessment feature. The baseline case confirmed the expected dynamics of an uncontrolled epidemic: the virus spreads rapidly, reaching its peak within the first days, and by the tenth day the entire population has been infected. This aggressive growth underlined the necessity of deploying effective countermeasures.

When applying only the 2D-Co-Contact algorithm, the outbreak curve was delayed compared to the uncontrolled scenario, and the number of daily infections remained lower during the early stages. Nevertheless, although the pandemic was slowed down, the virus eventually infected a significant amount of the population. The results showed that while contact tracing mitigates short-term spread, it was not sufficient to fully suppress the epidemic. The most notable improvements arise when Area Risk Assessment is introduced alongside the 2D-Co-Contact algorithm. By dynamically evaluating the riskiness of local zones and redirecting obedient agents away from critical areas, the spread is further slowed, and the peak of infections shifts later in time.

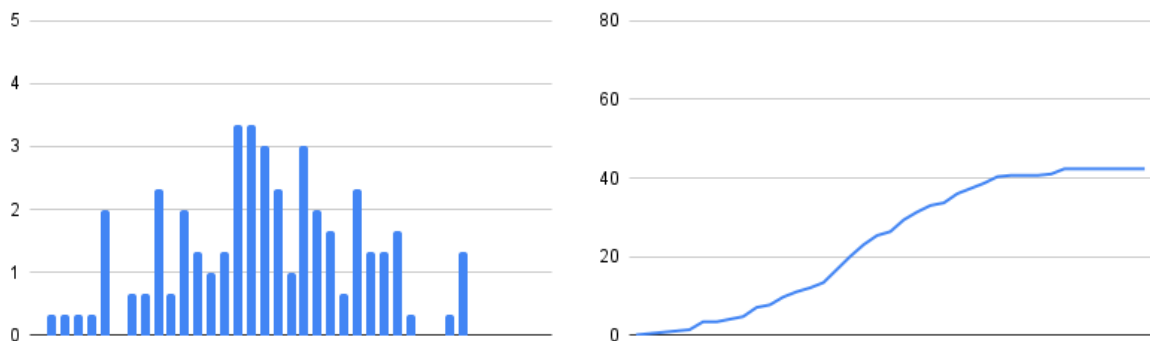


Figure 20. Daily new infections using the 2D-Co-Contact algorithm and the Area Risk Assessment with 20% obedient population and Cumulative infections graph.

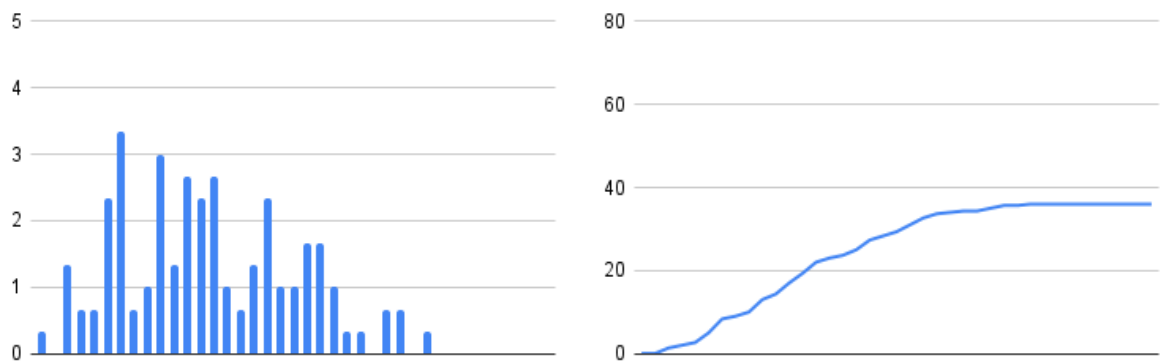


Figure 21. Daily new infections using the 2D-Co-Contact algorithm and the Area Risk Assessment with 40% obedient population and Cumulative infections graph.

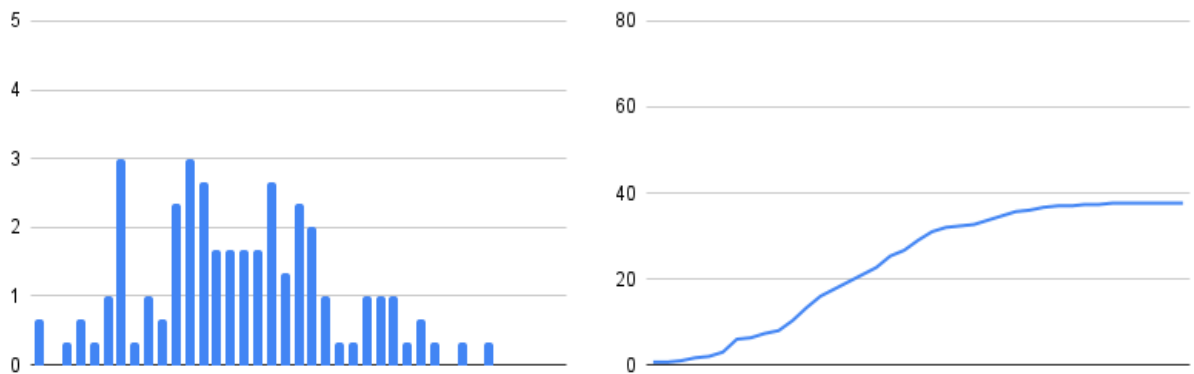


Figure 22. Daily new infections using the 2D-Co-Contact algorithm and the Area Risk Assessment with 60% obedient population and Cumulative infections graph.

At 20% obedience in Figure 20, we detect that the epidemic curve initially follows the trajectory of the plain 2D-Co-Contact algorithm, showing little difference during the early days of the simulation. However, once the first high-risk zones are detected and alerts are issued, the behavior diverges: daily infections begin to stabilize at lower levels, and the cumulative infection curve rises more gradually. This indicates that even limited compliance introduces an effective dampening of transmission, reducing the immediate growth pressure on the system.

At 30% and 40% obedience in Figures 17 and 21 respectively, the mitigation effect becomes clearer. The peak of daily infections is delayed further compared to both the baseline and the plain contact tracing model, and the total number of infected individuals over the 40-day horizon decreases noticeably. This suggests that as compliance rises, not only is the epidemic peak shifted to a later point in time, but the absolute scale of the outbreak is also diminished, leaving a larger share of the population uninfected.

With higher compliance levels of 60% in Figure 22, the differences are even more pronounced. Daily new cases remain consistently low throughout the experiment, and the epidemic peak is both substantially reduced and pushed to a much later stage. In these scenarios, a significant portion of the population remains uninfected by the end of the 40-day period, showing that area-based alerts combined with behavioral adaptation can spread to a much greater extent. Furthermore, the effective reproduction number R_0 declines more rapidly in these experiments, demonstrating that higher obedience

accelerates the disruption of transmission chains and reduces the chances of secondary outbreaks.

Taken together, the results illustrate a clear hierarchy of effectiveness. The uncontrolled environment produces a rapid and unavoidable pandemic, while the plain 2D-Co-Contact algorithm manages to slow the spread but ultimately fails to prevent large-scale transmission. In contrast, the integration of Area Risk Assessment significantly strengthens the system's capacity to mitigate the epidemic, particularly as the share of obedient agents increases. These findings validate the importance of coupling contact tracing algorithms with localized risk evaluation, as this combined approach provides both temporal delays and reductions in epidemic size, making pandemic prevention more feasible in complex urban environments.

Section 6. Conclusion

In this thesis, we proposed an integrated simulation framework for studying the spread of airborne pathogens in urban environments and evaluating the effectiveness of algorithmic countermeasures. The framework combined three core components—the spatial model, the mobility model, and the propagation model—providing a modular and extensible foundation for epidemic analysis. Building on this baseline, we introduced the Area Risk Assessment in Critical Zones, which enhances the propagation model by dynamically evaluating local exposure risk and guiding compliant agents to adapt their mobility. Through systematic experimentation, we demonstrated how varying levels of citizen obedience significantly affect epidemic outcomes, with higher compliance leading to delayed peaks, reduced daily infections, and lower cumulative cases. Comparisons against the baseline model and the plain 2D-Co-Contact algorithm further validated that the integration of localized risk awareness substantially improves epidemic mitigation. Overall, the work presented in this thesis highlights the importance of combining contact tracing with spatially informed risk evaluation, offering a more realistic and effective approach to pandemic prevention in complex urban settings.

6.1 Remarks

The findings of this thesis demonstrate that epidemic dynamics are deeply influenced not only by the biological properties of the pathogen but also by spatial, behavioral, and algorithmic factors that shape human interaction. By extending the baseline propagation framework with the proposed Area Risk Assessment procedure, we showed that localized risk evaluation, when combined with varying levels of compliance, can significantly reduce the impact of an outbreak and delay its progression. Also, via coupling this risk evaluation with an obedience factor, we simulated heterogeneous behavioral responses, where compliant agents reroute or return when entering high-risk zones. These findings confirm that the integration of contact tracing with area-based risk validation provides a more robust and effective strategy for pandemic mitigation, highlighting the critical role of spatial and behavioral modeling in public health preparedness.

6.2 Future Research

Following the completion of this work, future research will focus on extending the proposed framework beyond its current capabilities. Besides further optimizing the implemented simulation platform, we plan to incorporate dynamic navigation functions,

where citizens are guided to their destinations while actively avoiding congestion and high-risk areas, based on the real-time viral load of city routes derived from the movement traces of confirmed and potential infection cases. To achieve this, further research in algorithms, graph theory, and network optimization will be required to ensure efficiency and flexibility at scale. In alignment with the SARiSsa framework, another direction is the development of a mobile smart application that utilizes Bluetooth technology to log close contacts and provide area risk alerts in real time. Such an application would integrate both tracing and localized risk assessment, enabling the deployment of the most effective countermeasures quickly and efficiently. Ultimately, the long-term goal of this line of research is to deliver a scalable and adaptive decision-support tool that can help mitigate the spread of future airborne pathogens in complex urban environments.

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